Counting Shadows: A Community-Anchored Modeling Framework for Estimating Unauthorized Population Size*

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Abstract

Estimating the size of hidden populations typically requires linking individual records across multiple sources, such as by capture–recapture methods, which is often either impossible or error-prone. We estimate unauthorized foreign populations from aggregated administrative data alone, by proposing a general community-anchored modeling framework. Our model represents the unobserved unauthorized stock as a power-law function of an observed authorized reference population with the exponent as the community-anchor parameter, while the detection of unauthorized foreigners is regulated by an exposure-elasticity parameter, where both the parameters

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vary with covariates. Applied to Poland over 2019–2024, the method yields an estimated adult non-Schengen unauthorized population of 94,000 (80% interval: 36,000–206,000) in 2019 and 101,000 (46,000–177,000) in 2024, compared to the authorized reference population that has doubled over this period from 0.6 to 1.2 million. Moreover, we recover a structural break: the Ukrainian component declined from 60,000 to 11,000 following the 2022 EU Temporary Protection Directive, while the non-Ukrainian population expanded from 34,000 to 89,000. By removing the need for record linkage, our approach offers a viable alternative for many countries.

Keywords: Hidden populations, Aggregate data, Community-anchored models, Unauthorized migration, Official Statistics

1 Introduction

In 2025, YouGov conducted a survey in France, Germany, Italy, Spain, Denmark, Poland, and the UK, finding that support for reducing immigration is high and that these sentiments may be influenced by misperceptions about the scale of irregular migration. For example, in Poland, 36% believed that there are more unauthorized than lawful migrants, compared to 28% who think the opposite, and 22% who believe the numbers are roughly equal (YouGov, 2025). At the same time, close to 5 million migrants were registered in the Polish population register (PESEL) at the end of 2025, while those detected by Polish authorities as staying illegally numbered only about 10,000, though this figure captures only detected cases, not the total unauthorized population (Polish Border Guard, 2025). Pew Research Center (2019) estimated the unauthorized population of Poland at below 100,000 in 2014, 2015 and 2017 and at 100,000–200,000 in 2016.

This type of mistaken belief is not new and is driven by various factors (cf. Alesina et al., 2023; Ajzenman et al., 2023; Lutz and Bitschnau, 2023). Given that misperceptions stem from being misinformed and uninformed, plausible estimates of the hidden population size are essential for designing public policies, allocating public funding, and planning public services.

The size of the native-born population in a given country may be reasonably well-known at the national level, although the methods vary greatly across countries due to differences in statistical infrastructure: presence or absence of population registers (as in the UK and USA), variation in register quality, reliance on large-scale sample surveys (such as the American Community Survey; ACS) or population censuses. See e.g. Skinner (2018) or Dunne and Zhang (2024).

Foreign nationals can be generally divided into two main groups: those who stay or reside legally with valid permits, and those who are unauthorized—staying or residing without valid documentation. The latter group, in particular, is widely recognized as hard-to-reach and hard-to-measure (McKenzie and Mistiaen, 2009; Font and Méndez, 2013), including the countries that have high-quality population registers (Dańko et al., 2024; Monti et al., 2020; Careja and Bevelander, 2018), as by definition these people are excluded from the population register or other important registers, and also have incentives to avoid detection. In this paper, we focus on the unauthorized population.

Poland provides a case-in-point that illustrates the estimation challenge. First, between 2015 and 2024, the registered foreign nationals increased from 200,000 to nearly 4.5 million[1]. Second, between 2018 and 2023, Poland issued over 6 million visas[2] (about 3.8 million of which were work visas), placing it among the EU countries issuing the most visas of this type. Third, following Russia's invasion of Ukraine in February 2022, over 3.5 million refugees transited through Poland between February and mid-May 2022, with about 1 million remaining under temporary protection as of 31 December 2025 (cf. Statistics Poland (2025); Duszczyk and Kaczmarczyk (2022); Duszczyk et al. (2023), and https://ukraincywpolsce.stat.gov.pl). Finally, from 2021, Russia and Belarus have placed pressure on the Belarusian-Polish-EU border through politically instrumentalized migration, influencing the scale of unauthorized entry, but with limited political implications (Charasz and Menon, 2026). This combination of rapid demographic change, high visa issuance, humanitarian flows, and border pressures creates conditions where unauthorized populations are likely substantial

[1]These figures are based on the population (PESEL hereinafter) register and indicate the number of people who registered for a PESEL identifier, not how many actually reside in Poland. We discuss other registers later in the paper.

[2]This number refers to all visas types; it includes short-term visas, renewals, and all consular visa issuances. For more details see Supreme Audit Office of Poland (2024).

but hard to estimate.

In this paper, we propose a general modeling framework for estimating the size of the unauthorized foreign population, building on the community-anchored model introduced by Zhang (2008) for a one-off application to Norwegian data. Zhang (2008) established identifiability of the power-law relationship between authorized and unauthorized populations under Poisson log-linearity; we extend it into a framework that accommodates data over multiple years and subpopulations, with covariate-varying parameters, robust inference of uncertainty, and estimator choice guided by simulation evidence.

Unlike other existing approaches — such as the residual method (cf. Warren and Passel, 1987; Chassamboulli and Peri, 2015; Van Hook et al., 2021; Warren et al., 2025), capture-recapture or multiple systems estimation (cf. Chipperfield et al., 2024; Böhning and Friedl, 2024; Aleshin-Guendel et al., 2024), and deductive or digital-trace methods (Recchi et al., 2026) — which require linkage across multiple data sources or unit-level micro data at specific time points, our method requires only aggregated data. We develop a model grounded in demographic and behavioral assumptions that uses three inputs: *aggregated counts* of detected unauthorized migrants, the size of the authorized foreign nationals, and a covariate proxying for detection intensity.

Our key assumptions include: (1) there exists a structural relationship between unauthorized and authorized foreign nationals due to the former's reliance on communities or networks for survival; (2) it is valid to target a theoretical size of unauthorized population without spurious day-to-day variations; and (3) the detection rate of unauthorized foreigners can be modeled statistically as a function of known features that proxy the relevant channels or mechanisms.

Support of such a theoretical model can be found in two earlier EU-funded research programs. The CLANDESTINO project (CLANDESTINO Consortium, 2009) documented the extent to which unauthorized migrants depend on established co-ethnic communities for housing, employment, and social support — the same network mechanism is captured by the community anchor parameter in our model. The MIrreM project (Kierans and Kraler, 2025) highlighted severe biases of sizing

by raw apprehension data directly, which reflect enforcement priorities and capacity constraints rather than the underlying unauthorized population — the insight is reflected in our detection intensity parameter which connects to the observed counts of unauthorized people to the unobserved population sizes. Our modeling framework thus answers a call form both these projects for methods that move beyond the contested administrative counts toward credible population estimates.

Applied to the Polish data from 2019 to 2024, our approach yields plausible trajectory of estimates despite the structural shock caused by the Ukraine war. Our estimate of the adult (18+) non-Schengen unauthorized population size in Poland stands at about 94,000 (80% interval: 36,000–206,000) in 2019 and 101,000 (80% interval: 46,000–177,000) in 2024.[3] A central message is one of relative decline: even as Poland's registered foreign population surged and the Social Insurance Institution (ZUS) reference population almost doubled (from 612,000 to 1,134,000), the estimated unauthorized population remained broadly stable, so its share of the ZUS reference population fell from 15.4% in 2019 to 8.9% in 2024. This relative finding is robust to the considerable width of the underlying confidence intervals.

Moreover, our estimates capture divergent dynamics by the country of origin. The estimated Ukrainian unauthorized population declines from 60,000 in 2019 to 11,000 in 2024, consistent with the mass regularization under the 2022 EU Temporary Protection Directive, while the non-Ukrainian component moves in the opposite direction, rising from 34,000 to 89,000 over the same period.

Sensitivity analyses suggest that these estimates are robust against model mis-specifications that are unavoidable, and aligns plausibly with other administrative data used in the estimation. In particular, the wide confidence intervals reflect the inherent uncertainty in estimating hidden populations (cf. Van Hook et al., 2021), and it is important for users and policy makers to pay attention to the range of plausible values rather than relying too much on any single point estimate.

Finally, our approach is easily transferable to other countries, since it relies only on aggregated

[3] In the abstract and introduction we report 80% intervals; the tables report both 80% and 95% intervals. Because the bootstrap distribution is right-skewed, the interval bounds indicate estimation uncertainty rather than equally plausible population sizes, and the central (median) estimate is the most probable value. Point estimates and interval bounds for unauthorized population sizes are rounded to the nearest 1,000; detailed tables retain finer precision.

administrative data that exist commonly, and its efficacy has already been demonstrated in quite different situations from Norway to Poland. The relevant methodology has the potential for National Statistical Offices to improve migration estimates by adding the *uncounted* component.

In the rest of the paper, Section 2 presents the theoretical model, Section 3 the empirical specification, Sections 4–5 describe the application to Polish data, and the conclusions are drawn in Section 6. Additional details are given in the Supplementary Materials. The open-source R (R Core Team, 2026) package `uncounted` is available at https://github.com/ncn-foreigners/uncounted. Replication materials are available here https://github.com/ncn-foreigners/paper-shadows.

2 Theoretical model

To formalize our theoretical model, let M_{it} be the size of the unauthorized population of *community*[4] i $(i = 1, \ldots, I)$ in period t $(t = 1, \ldots, T)$, and let N_{it} denote the size of a corresponding authorized *reference population*. Let m_{it} be the *detected* count of unauthorized migrants (among the M_{it}), and let n_{it} denote a known count which may or may not correspond to a subset among the N_{it}, such that n_{it}/N_{it} provides a feature for *detection-intensity* related to m_{it}/M_{it}.

We define the *theoretical* size of unauthorized population as the conditional expectation of M_{it},

$$\xi_{it} = \mathbb{E}(M_{it} \mid N_{it}). \tag{1}$$

Let the conditional expectation of the *detection rate* $p_{it} = m_{it}/M_{it}$ given the others be

$$\rho_{it} = \mathbb{E}(p_{it} \mid M_{it}, N_{it}, n_{it}). \tag{2}$$

Combining (1) and (2) under the assumptions (i) M_{it} and p_{it} are conditionally independent given (N_{it}, n_{it}), and (ii) n_{it} provides no additional information about M_{it} beyond N_{it}, we obtain the

[4]Note that we use the term community to indicate that our approach can also be applied where data at the country-of-origin level are not available (e.g. countries are grouped by ethnicity).

expectation of the detected count $m_{it} = M_{it}p_{it}$ as

$$\mu_{it} = \mathbb{E}\{M_{it} \cdot \mathbb{E}(p_{it} \mid M_{it}) \mid N_{it}, n_{it}\} = \mathbb{E}(M_{it} \mid N_{it}) \cdot \mathbb{E}(p_{it} \mid N_{it}, n_{it}) = \xi_{it} \cdot \rho_{it} \tag{3}$$

The functional forms of ξ_{it} and ρ_{it}, as well as the distributional assumptions of m_{it}, will be treated later as part of our empirical model in Section 3. Below we discuss first the motivation and justification of the theoretical model defined by (1)–(3).

First, our core assumption is that the size of the unauthorized population is structurally related to the authorized population via communities, which typically concur with the country of origin, ethnicity, or similar characteristics. This is supported by migration research showing that unauthorized migrants depend heavily on co-ethnic and kin-based networks for survival and incorporation into host societies (Hagan, 1998; Engbersen and van der Leun, 2001; Ambrosini and Hajer, 2023). Such networks serve multiple functions: providing accommodation, information, informal employment, and emotional support, and often substitute for formal welfare and labour market institutions that unauthorized migrants cannot access (Anderson, 2007; McDowell et al., 2009). Consequently, we expect the unauthorized population from a given origin country to scale with the corresponding authorized population, which serves as the foundation for these support networks.

Moreover, unlike the residual or capture-recapture methods, which treat the unauthorized population size as an *unknown* constant, we view M_{it} as a *random* variable arsing from processes driven by economic, political, and demographic factors, and model its theoretical size ξ_{it} in relation to a suitable reference population size N_{it}, i.e. Equation (1). This allows us to capture how the two populations co-evolve: as authorized communities grow, they provide expanded networks that support larger unauthorized populations from the same origin or ethnicity. Conditioning on N_{it} also makes it possible to interpret the estimate of ξ_{it} without spurious day-to-day variations, just like how standard official population statistics are received.

Regarding the detection rate p_{it} that connects the observed count m_{it} and the unobserved population size M_{it}, the factorization in (3) requires that $\mathbb{E}(p_{it} \mid M_{it}, N_{it}, n_{it}) = \mathbb{E}(p_{it} \mid N_{it}, n_{it})$, meaning the

detection rate does not depend on the size of the unauthorized population. This assumption rules out feedback effects whereby detection intensity ρ_{it} responds endogenously to M_{it}. Although it cannot be tested directly since M_{it} is unobserved, the data of staffing and enforcement activities in Poland (see Table S7 in Supplementary Materials) show that legality-of-stay inspections fluctuated over 2019–2024 without a systematic trend, Police and Border Guard staffing remained broadly stable despite substantial growth of the presence of foreign nationals, whereas employment legality inspections actually declined over the period. Altogether they suggest that detection intensity is shaped by supply-side factors (budget cycles, staffing constraints, pandemic disruptions) rather than by endogenous responses to M_{it}, rendering the no-feedback assumption empirically plausible.

Of course, we must caution against taking this for granted. For example, congestion effects may arise when enforcement resources are fixed relative to the unauthorized population, causing the detection rate to decline as M_{it} increases (Fender, 1999; Hanson and Spilimbergo, 1999). Conversely, concentration effects may cause detection rate to increase with M_{it} if larger populations are more visible to enforcement (Ethier, 1986). However, without additional data or instrumental variables, it is infeasible to distinguish the variation in p_{it} from that in M_{it} when the former depends on the latter (Link, 2003). In other words, different approaches will be necessary in such situations.

Finally, by way of introduction to the empirical model, we notice the flexibility of this theoretical model. On the one hand, it allows one to estimate the total size of unauthorized population without decomposing the counts m_{it} into visa overstayers, border apprehensions, and so on. Attempts to deal with these subgroups separately, originated from different data sources, may cause complications and aggregation inconsistencies (Warren and Passel, 1987). On the other hand, it is possible to disaggregate m_{it} by relevant covariates such as gender and age categories, when it comes to the empirical model specification, provided this is sensible and supported by the available data.

3 Empirical model

3.1 Parametric specification

As the first part of our empirical model, we now specify the functional forms for (1) and (2). For the theoretical size of unauthorized population, a power-law relationship is given as

$$\xi_{it} = \mathbb{E}(M_{it} \mid N_{it}) = N_{it}^{\alpha_{it}}, \tag{4}$$

where $\alpha_{it} > 0$ is an unknown *Community Anchor* parameter, capturing how the unauthorized population scales with the reference population for the given community and time. Let the corresponding detection intensity ρ_{it} be given as

$$\rho_{it} = \mathbb{E}(p_{it} \mid N_{it}, n_{it}) = \left(\gamma + \frac{n_{it}}{N_{it}}\right)^{\beta_{it}}, \tag{5}$$

where $\beta_{it} > 0$ is an unknown parameter of *Exposure Elasticity*, regulating how the detection intensity responds to the chosen feature n_{it}/N_{it}, and $\gamma > 0$ represents a *(baseline) offset* even when $n_{it} = 0$. Notice that $\rho_{it} \in (0, 1)$ as long as $0 < \gamma + n_{it}/N_{it} < 1$.

The community-anchored model of Zhang (2008) is a special case that uses the simplest parameterization $(\alpha_{it}, \beta_{it}, \gamma) \equiv (\alpha, \beta, 0)$ for a one-off application in Norway, where t refers to a single year 2006 and i refers to the countries of citizenship. In particular, all the countries with $m_i n_i = 0$ are *regrouped* into a 'rest' community, so that $\rho_{it} > 0$ everywhere even with $\gamma \equiv 0$. The parameterization here in terms of (4) and (5) is more general and principled.

Firstly, to accommodate multiple periods and flexible formulation of communities, we allow α_{it} to vary with (x_i, t), where x_i denotes the relevant covariates of the communities. For example, let $x_i = 1$ for males and $x_i = 2$ for females from country i. By suitable dummy-coding of (x_i, t), we can allow distinct α-values from crossing x_i and t directly, or as the sum of a gender effect and a year effect but without any interaction of the two. Similarly for β_{it}, with a separate parameterization,

such as $\beta(t)$ and $\alpha(x_i, t)$ at the same time.

Next, conceptually speaking, the detection intensity ρ_{it} is never zero since, even when the feature count is $n_{it} = 0$, it is still possible to detect unauthorized migrants through various channels, such as border guards, hospital admissions, rejected visa or asylum applications, employer inspections. Introducing the offset γ is a more principled approach than regrouping to form ad hoc 'rest' communities, because the composition of such 'rest' communities will not be the same over time. Moreover, it allows one to make better use of the data points with $m_{it} > 0$ and $n_{it} = 0$.

3.2 Motivation and justification of the parametric specification

Various functional forms could relate ξ_{it} to N_{it}. Our choice of the power-law specification (4) is consistent with the considerations underlying the theoretical assumption (1). Unauthorized migrants lack access to regular employment and welfare services, requiring a network of contacts to survive (Hagan, 1998; Engbersen and van der Leun, 2001). These contacts most likely include usual residents in established socio-economic situations, which provides the "existence basis" for the unauthorized population, analogous to how a reproductive population supports offspring in demographic models.

It is therefore natural to expect a relationship where ξ_{it} increases with N_{it}, but the rate of increase declines as N_{it} becomes larger[5]. In particular, our specification closely relates to the Beverton and Holt (1957) model, which is a discrete-time population model given by

$$R = \frac{\nu S}{1 + \kappa S},$$

where S is the current population size, R is the next-generation population, κ is the density dependence parameter, and ν/κ is the carrying capacity. The power curve N^α and the Beverton-

[5] Massey (1990) introduced the idea of cumulative causation in migration, emphasizing how migrant networks create self-reinforcing migration processes. Fussell and Massey (2004) show that these mechanisms are much weaker in large urban settings than in smaller communities. Carrington et al. (1996) provide a model in which moving costs decline as migrant networks expand, generating gradual self-reinforcing migration dynamics.

Holt curve exhibit similar density-dependent behavior, particularly over empirically relevant ranges of N. The key difference is that the Beverton-Holt model assumes an asymptotic ceiling ν/κ, whereas the power-law specification allows $\xi_{it} \to \infty$ as $N_{it} \to \infty$. An absolute ceiling of the unauthorized population size is not necessary in our context.

In contrast, the power-law specification (5) does not suggest a direct connection from the chosen feature rate n_{it}/N_{it} to the detection rate m_{it}/M_{it} of unauthorized migrants. The count n_{it} should be chosen such that the corresponding rate n_{it}/N_{it} may proxy for detection intensity through various mechanisms, the elasticity of which can be tuned by the parameter β_{it}. For instance, if n_{it} is based on police records of foreign nationals facing criminal charges, there is no direct connection between criminal behavior and unauthorized status—rather, contact with law enforcement creates opportunities for detection regardless of criminality. Similarly, when n_{it} is derived from events recorded in administrative systems, it may proxy the degree to which migrants interact with the relevant institutions or agencies that may detect unauthorized foreigners (e.g., hospital admissions, rejections of visa/asylum applications from the previous year).

We notice that it is possible to adopt logistic functions to ensure $\alpha_{it} < 1$ or $\rho_{it} < 1$; see Supplementary Materials Section H. However, log transformation corresponding to the power-specification is preferable because it enables us to combine equations (4) and (5) directly on the log scale, as long as the restrictions are empirically unnecessary for the data at hand.

We obtain straightforward model diagnostics as one of the benefits. Plugging in m_{it} for $\mu_{it} = \xi_{it}\rho_{it}$ given by (4) and (5), dividing both sides by N_{it} and taking logarithms for the data points with $n_{it} > 0$ yields, for each period t with $(\alpha_{it}, \beta_{it}, \gamma) = (\alpha_t, \beta_t, 0)$,

$$\log\left(\frac{m_{it}}{N_{it}}\right) = (\alpha_t - 1)\log N_{it} + \beta_t \log\left(\frac{n_{it}}{N_{it}}\right). \tag{6}$$

Plotting $\log(m_{it}/N_{it})$ against each covariate on the right side provides a diagnostic check: by the considerations above, we expect $\alpha_t - 1 < 0$, i.e., a negative relationship with $\log N_{it}$, and $\beta_t > 0$, i.e., a positive relationship with $\log(n_{it}/N_{it})$. Furthermore, stratifying these plots by country

groups (e.g., by visa requirements, continent or migration corridor) allows us to assess whether a single pattern holds uniformly or systematic differences exist to warrant more disaggregated parameterization. Similarly with other partitions of all the data points for exploratory data analysis.

Finally, although the model can be tested empirically for goodness-of-fit to observed counts m_{it}, the specifications (4) and (5) cannot be separately verified. In other words, it is logically possible for other empirical models of μ_{it} to fit equally well to the observed counts m_{it}, such that a choice must rely on certain considerations that cannot be directly tested. This is a fundamental limitation inherent to estimating hidden populations, and we would caution against overconfidence in the results. Nevertheless, we have not found decisive evidence against specifications (4) and (5) in our applications, and the motivations outlined above provide a coherent rationale.

3.3 Parameter estimation

To estimate the mean structure (4)–(5), it suffices to assume independent Poisson distributions of m_{it}, denoted by

$$m_{it} \sim \text{Poisson}(\mu_{it}), \quad \log(\mu_{it}) = \alpha_{it}(\eta) \log N_{it} + \beta_{it}(\eta) \log\left(\gamma + \frac{n_{it}}{N_{it}}\right), \tag{7}$$

where η contains all the unknown constants in the specifications of $(\alpha_{it}, \beta_{it})$, and $\Pr(m_{it} = 0) > 0$ exists by the Poisson distribution such that all the data points can be used in the estimation directly. However, since the variance structure of m_{it} is most likely misspecified in this way, we shall refer to the resulting estimator as a pseudo maximum likelihood estimator (PMLE), instead of MLE given the Poisson distribution. The Poisson PMLE would coincide with the MLE when the Poisson distribution holds, but it requires only correct mean specification for consistency regardless of the true variance structure (Santos Silva and Tenreyro, 2006).

Notice that Zhang (2008) accommodates over-dispersion of m_{it} by explicitly assuming

$$m_{it} \sim \text{Poisson}(\mu_{it} u_{it}) \quad \text{and} \quad u_{it} \sim \text{Gamma}(1, \phi),$$

where Gamma$(1, \phi)$ denotes the gamma distribution with expectation 1 and variance $1/\phi$. The Poisson-Gamma mixture yields the Negative Binomial (NB) likelihood on integrating out the unobserved random effects u_{it}, and the corresponding NB-MLE. In Norway applications, the NB-MLE was obtained with $\gamma \equiv 0$ in (5) and regrouping of data points to avoid $m_i n_i = 0$.

Given non-trivial offset $\gamma > 0$, the NB-MLE is feasible without regrouping data points. Due to its ability to accommodate over-dispersion of m_{it}, which exists commonly in real data, the formal model-selection criteria such as AIC or BIC tend to favor NB-MLE to Poisson PMLE. However, as we explain in Supplementary Materials D.1 that there are several reasons for preferring the Poisson PMLE to NB-MLE in practice, specially when non-trivial offset $\gamma > 0$ is allowed in (5).

3.4 Estimation of unauthorized population size

Given the estimates $\hat{\alpha}_{it} = \alpha_{it}(\hat{\eta})$ from (7), we can estimate the overall size of unauthorized population for a period t as

$$\hat{\xi}_t = \sum_{i=1}^{I} N_{it}^{\hat{\alpha}_{it}} \tag{8}$$

Consistency of $\hat{\xi}_t$ follows from the Continuous Mapping Theorem applied to $g(\hat{\eta})$. Meanwhile, in finite-sample applications, one may consider to reduce the potential bias due to the non-linear transformation involved, which would result in an estimator given as

$$\hat{\xi}_t^{\text{BC}} = \sum_{i=1}^{I} N_{it}^{\hat{\alpha}_{it}} \exp\left(-\frac{1}{2}(\log N_{it})^2 \sigma_{it}^2\right), \tag{9}$$

where σ_{it}^2 is a term related to the variance of $\hat{\eta}$ — see Supplementary Materials Section H for the full derivation and Supplementary Materials Section F for finite-sample simulation results. However, the adjustment is typically small compared to the width of confidence interval.

A confidence interval of α_{it}, denoted by $(\hat{\alpha}_{it}^{\text{lower}}, \hat{\alpha}_{it}^{\text{upper}})$, can be obtained using heteroskedasticity-consistent (HC) standard errors (MacKinnon and White, 1985). Since $g(\alpha) = \sum_{i=1}^{I} N_{it}^{\alpha}$ is mono-

tonically increasing in α given $N_{it} \geq 1$, a confidence level of ξ_t is given directly as

$$\left(\sum_{i=1}^{I} N_{it}^{\hat{\alpha}_{it}^{\text{lower}}}, \sum_{i=1}^{I} N_{it}^{\hat{\alpha}_{it}^{\text{upper}}} \right). \tag{10}$$

However, in practice, one may still encounter situations where the interval can be excessively wide due to ill-behaving $(\hat{\alpha}_{it}^{\text{lower}}, \hat{\alpha}_{it}^{\text{upper}})$ for a subset of all α_{it}.

We therefore prefer the fractional weighted bootstrap [FWB; Xu et al. (2020)] with clustering by country-of-origin. The approach simultaneously accounts for within-country correlation across years and covariates, the nonlinear transformation of parameter estimates, and their skewed distributions. In particular, one may consider the bootstrap median of $\{\hat{\xi}_t\}$ as the point estimate, which is more robust to skewness than either a plug-in estimator or the bootstrap mean.

It is worth noting that, in applications, wide confidence intervals are a well-documented feature of unauthorized population estimation, reflecting the inherent uncertainty in estimating hidden populations. This has been discussed extensively in the migration literature, including Van Hook et al. (2021), the Centre on Migration, Policy and Society (COMPAS) estimates at the University of Oxford[6], the CLANDESTINO project (cf. CLANDESTINO Consortium, 2009), and the MIrreM project (Kierans and Kraler, 2025).

4 Data sources

4.1 Definition of an unauthorized stay

In our study, we use Polish Border Guard reports, which distinguish three groups of migrants: (G1) refusals of entry at the border, (G2) detections during irregular border crossings, and (G3) detections for illegal stay within the country or at the border when leaving. We focus on the third group, and in particular those who were identified *within the country*. Groups G1 and G2 capture border-level encounters (attempted or actual irregular crossings), whereas group G3 reflects

[6] See https://migrationobservatory.ox.ac.uk/resources/briefings/unauthorized-migration-in-the-uk/

detections of unauthorized migrants already present on Polish territory, which corresponds to the target population M_{it} in our theoretical model. Within group G3, we exclude those detected at the border upon departure, as they are in the process of leaving the unauthorized population and their inclusion could conflate exit flows with the resident stock. Detailed definitions of each group are provided in Supplementary Materials Section A.1.

Under Polish law, an unauthorized stay encompasses any presence that violates entry or residence regulations—ranging from expired visas and overstayed permits to illegal employment and Schengen Information System flagged individuals. Two authorities verify legality of stay: the Border Guard (at the border and within the country) and the Police. Note that a person who crossed the border illegally but was subsequently detected within the country falls into our analysis sample under group G3. For the full legal definition, consequences, and recent legislative changes, see Supplementary Materials Section A.1 and A.2. For a comprehensive overview of the forms and drivers of migrant irregularity in Poland—including visa overstaying, fictitious employment declarations, and expired pandemic-era permits—see Yeliseyeu et al. (2024). Finally, The Police record foreigners mainly in pre-trial criminal proceedings, which cover offences prosecuted *ex officio* and not unauthorized stay, a petty offence handled by the Border Guard; the two sources therefore overlap only marginally (see Section A.3).

4.2 Data sources

4.2.1 Observed unauthorized population

The first dataset, on the number of unauthorized migrants detected for illegal stay, was prepared by representatives of the Polish Border Guards on a half-year basis with information on people who were identified multiple times, along with their country of citizenship, sex and age group. In our analysis we use counts of unique individuals.

Figure 1 presents the number of observed unauthorized migrants by place of apprehension by the Polish Border Guard for the period 2014–2024, aggregated by half-year periods. The observed

pattern is dominated by apprehensions at the Polish-Ukrainian border, which rose sharply from around 1,500 in early 2014 to a peak exceeding 10,000 in the first half of 2018, before declining after 2020 to levels comparable to those at other borders. This surge likely reflects the aftermath of the 2014 conflict in eastern Ukraine and subsequent economic migration pressures. Interior apprehensions ("Interior") show a steadier upward trend, becoming the dominant category by 2023–2024, suggesting a shift in migration patterns or enforcement priorities toward identifying unauthorized migrants already within Poland rather than at exit.

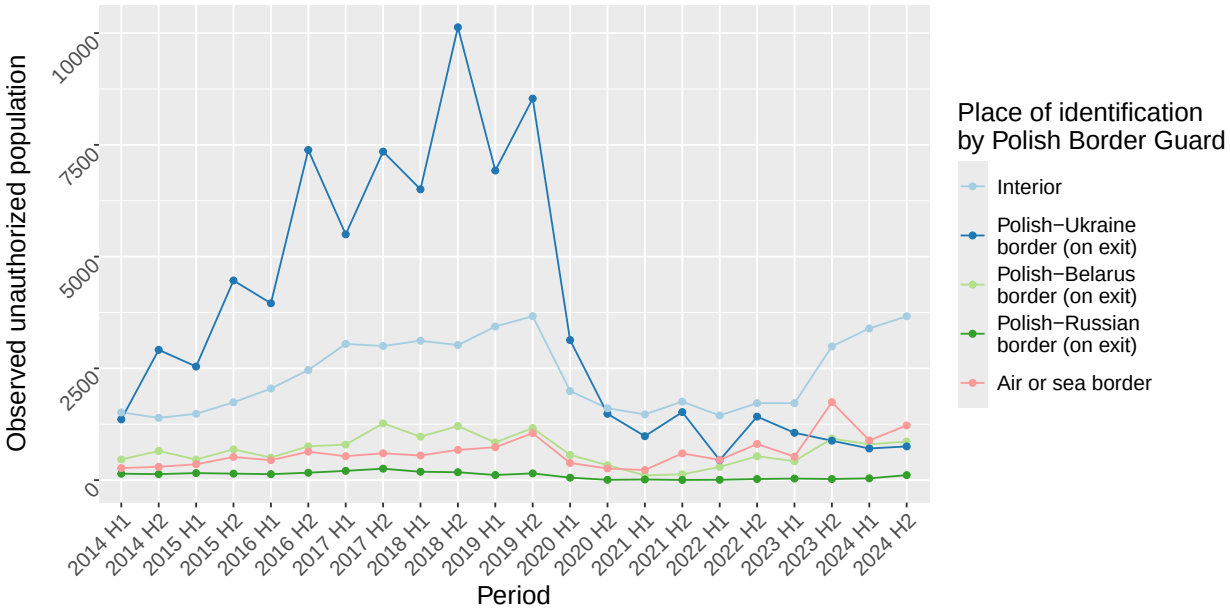


Figure 1: Number of unauthorized migrants aged 18+ apprehended by the Polish Border Guard, by place of apprehension, 2014–2024, aggregated by half-year periods. These data refer to individuals detained or detected by the Border Guard for illegal stay. Identification at the border refers to foreigners identified as unauthorized when leaving Poland (not on legal or illegal entry to Poland).

The Belarusian border shows modest increases around 2021–2022, possibly reflecting the well-documented migrant crisis orchestrated by Russian and Belarusian authorities during that period. Apprehensions at the Russian border and via air/sea routes remain consistently low throughout the entire period. The COVID-19 pandemic (2020) coincides with a sharp drop across all categories. After the Russian invasion of Ukraine in 2022, apprehensions at exit from Poland fell further and the Ukrainian border never regained its earlier prominence, as Ukrainians did not return to Ukraine

during the war, while interior identifications continued to grow. In the paper, we use *interior* unauthorized counts as our m.

4.2.2 Lawful foreign population counts

The second dataset refers to the reference, lawful population. Poland's population register (PESEL) reliably records the assignment of personal identifiers but not actual residence or stay, as de-registration is rarely reported—evident in Figure 2, where PESEL shows the highest counts (peaking at 4.1 million in 2024) but limited decline thereafter despite likely outflows[7].

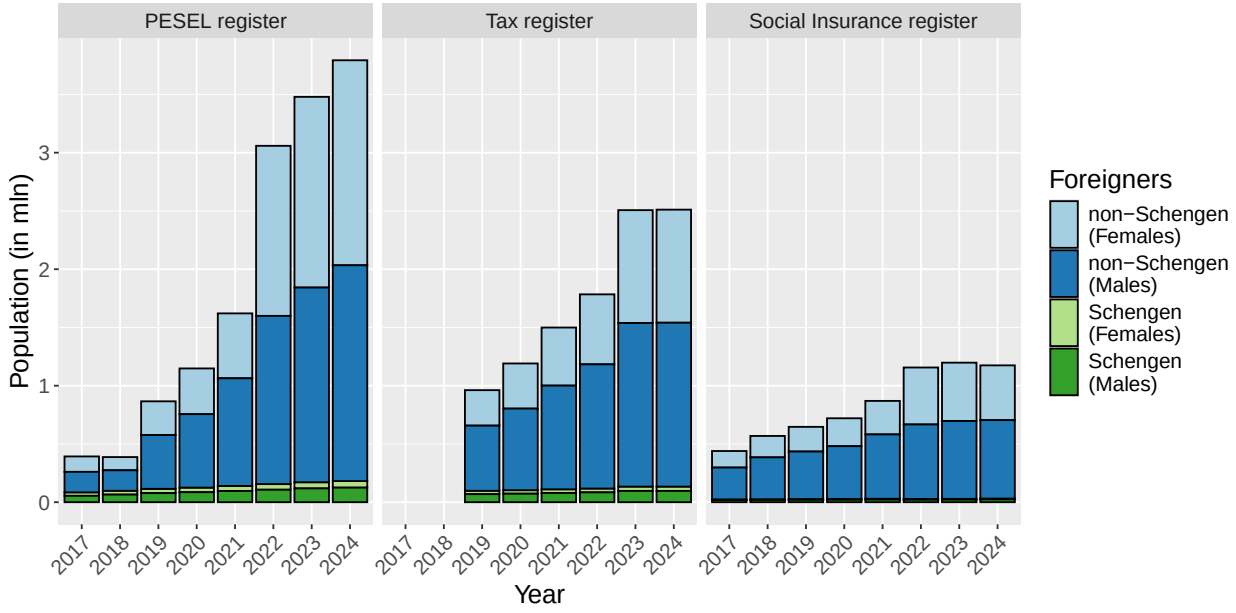


Figure 2: Number of 18+ foreign nationals by register, country of citizenship (Schengen, non-Schengen) and sex. Data from the Tax register were available only for the 2019–2024 period. All reference populations are as of the end of these years.

We therefore also considered Tax and Social Insurance (ZUS) registers, which better capture economic presence and show more modest figures (about 2.5 million and 1.2 million by 2023, respectively). For 2024, the number of foreigners in ZUS and Tax registers stagnated. Across all

[7] Górny and Kaczmarczyk (2023) report mobile phone estimates of 3.2–3.8 million Ukrainians in Poland in spring 2022 versus about 1.5 million PESEL UKR (special protection) registrations at that time. Strzelecki et al. (2025) estimate that about 1 million Ukrainian refugees settled in Poland by 2024 on top of 1.5 million pre-war Ukrainian immigrants and 0.9 million from other countries, contributing about 0.5 percentage points to annual GDP growth.

registers, non-Schengen nationals (predominantly Ukrainians) dominate and increased sharply after 2021, reflecting both labour migration and the post-February 2022 refugee influx. The concurrent shift from male to female predominance among non-Schengen migrants mirrors the demographic composition of war refugees.

4.2.3 Detection covariate and the choice of the reference population

The third source provides the feature for detection intensity. We use anonymised individual-level data from the Polish Police's National Police Information System (KSIP), which records foreign nationals who came into contact with law enforcement—including background information on sex, age, place of residence (if provided), and PESEL identifier. Our covariate covers process registrations and criminal records, i.e. suspects and accused persons in offences prosecuted *ex officio*. These records are aggregated by country of origin, year, and sex to produce the covariate n_{it} used in estimation. Contact with police does not imply unauthorised status; rather, it proxies the degree to which a country's migrants are visible to institutional detection mechanisms. KSIP has no unauthorised-stay ground for registration and holds no information on legality of stay (cf. Section A.3). Supplementary Materials Section D.6 refits the final model, discussed on the next section, with two alternative counts: police records restricted to persons holding a PESEL identifier, who are registered residents and thus least likely to be unauthorized, and prison records. Poisson estimates under both covariates stay within the intervals obtained with the full police count.

4.2.4 Summary of the datasets used in the study

A summary of the datasets used in this study is presented in Table 1. A detailed summary of zero-counts in terms of N_{it}, m_{it} or n_{it} can be found in the Supplementary Materials (Table S8 and S9). As can be seen, large differences exist between males and females in all the datasets.

The choice of reference population counts N affects primarily the community-anchor parameter α and secondarily the exposure-elasticity parameter β. We decided to use the Social Insurance Institution (ZUS) register as the source of reference population counts N for several reasons. First,

this register is updated monthly, providing timely information on the foreign nationals. Second, it contains records of foreign individuals registered for social insurance by their employers, which provides a strong signal of actual presence; since registration implies active employment, we can be reasonably certain that these foreigners are indeed residing, working or insured in Poland. In contrast, the Tax register is updated only annually, limiting its temporal resolution. Furthermore, the PESEL register suffers from well-documented deregistration issues, where individuals who have left the country may remain in the database, leading to overestimation of the resident population.

Table 1: Yearly information on the observed unauthorised, auxiliary and reference 18+ populations for non-Schengen countries between 2019 and 2024 by sex (in thousands)

				Register		
Year	Sex	Border[a]	Police[b]	PESEL[c]	Tax[c]	ZUS[c]
2019	Females	1.5	0.7	281.5	296.6	208.3
	Males	5.1	9.1	441.7	535.9	400.7
	Total	*6.6*	*9.8*	*723.1*	*832.5*	*609.0*
2020	Females	0.7	0.7	383.2	379.3	236.4
	Males	2.7	8.7	607.9	675.1	445.3
	Total	*3.4*	*9.4*	*991.0*	*1,054.3*	*681.6*
2021	Females	0.5	0.9	549.0	491.4	284.6
	Males	2.6	12.0	907.0	870.3	546.3
	Total	*3.1*	*12.9*	*1,456.0*	*1,361.7*	*830.9*
2022	Females	0.3	1.4	1,450.9	593.3	485.6
	Males	2.6	15.3	1,424.0	1,044.9	633.1
	Total	*3.0*	*16.7*	*2,875.0*	*1,638.2*	*1,118.7*
2023	Females	0.5	1.8	1,627.4	960.9	497.3
	Males	3.8	16.2	1,648.5	1,380.2	662.9
	Total	*4.4*	*18.0*	*3,275.9*	*2,341.1*	*1,160.2*
2024	Females	1.0	2.0	1,749.0	962.5	467.1
	Males	5.8	15.6	1,825.6	1,383.0	666.0
	Total	*6.7*	*17.6*	*3,574.5*	*2,345.5*	*1,133.0*

[a]Observed unauthorised population according to Polish Border Guard inspections within the country (excluding airports); this is m in our model.

[b]Auxiliary population used to model the detection rate ρ; this is n in our model.

[c]Reference population registers; serve as N in our model. Data recorded as of 31 December of each study year.

Data from Border Guard and Police are counts of persons within the year, while reference populations are stocks as of year-end.

5 Results

5.1 Exploratory analysis and estimation

We use diagnostic plots to explore directly the two key empirical model assumptions: a sub-proportional scaling of unauthorized population with the reference (ZUS) population (community anchor $\alpha < 1$) and a positive elasticity to the auxiliary covariate (exposure elasticity $\beta > 0$).

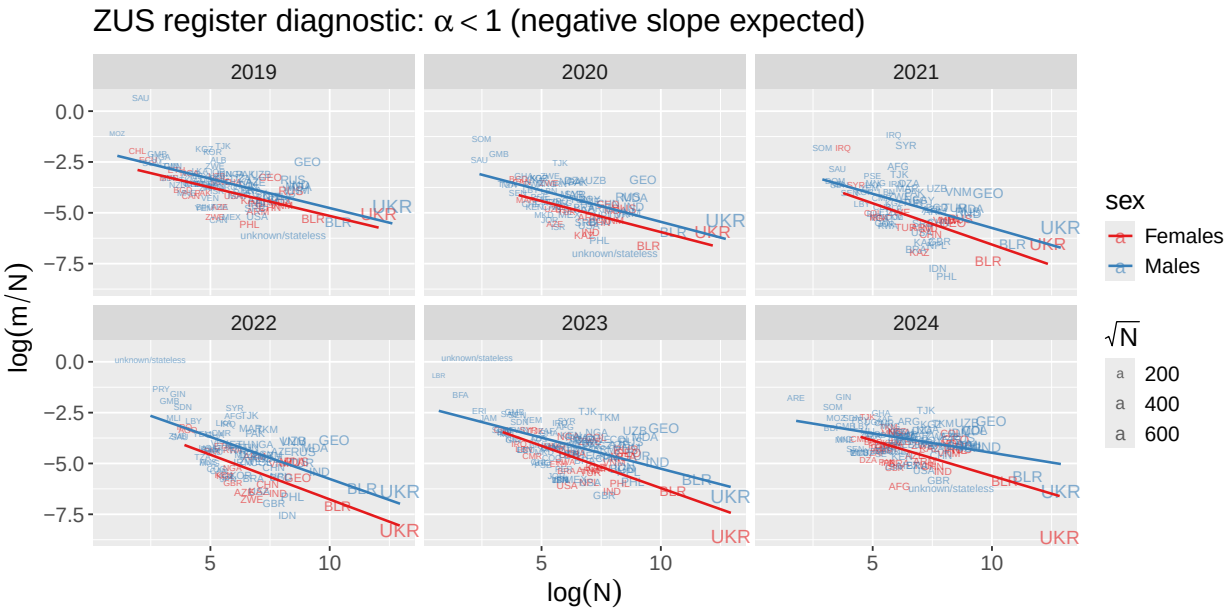


Figure 3: Relation between standardised positive detection count m/N and reference (ZUS) population size N on log scale. Size of country codes proportional to N, regression lines separately by (year, sex).

Figure 3 shows the relationship between $\log(m/N)$ and $\log(N)$, as explained for (6) earlier. Separate regression lines by (year, sex) are added to enhance the impression. The consistently negative slopes confirm the anticipated sub-proportional scaling above, which indicates that the community-anchor parameter takes a value between 0.5 to 0.8.

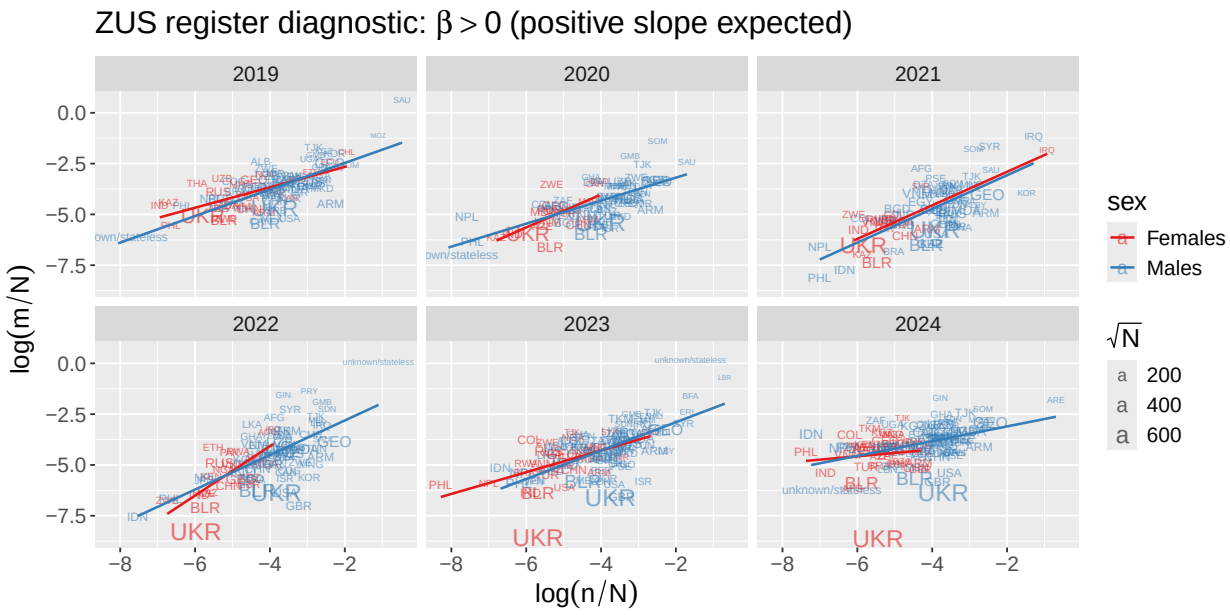


Figure 4: Relation between standardised positive detection count m/N and detection feature n/N on log scale. Size of country codes proportional to N, regression lines separately by (year, sex)

Similarly, Figure 4 presents the relationship between $\log(m/N)$ and $\log(n/N)$, including separate regression lines. The positive slopes confirm the anticipated positivity of exposure-elasticity parameter β. Their angles are close to 45 degrees in most cases, with values of β close to 0.8 up to 2023 with a decrease to about 0.4 in 2024, which shows that the police records can provide useful feature counts n in this application.

The diagnostics above are stratified by year and sex. Supplementary Materials Section C contains similar diagnostic plots for all three possible sources of reference population (ZUS, Tax, PESEL), as well as stratification by continent, which confirm that the two key empirical model assumptions hold broadly for alternative choices of N and partitions of communities.

5.2 Parameter specification and estimation

Of the parameters in the mean structure defined by (4) and (5), the community-anchor α_{it} affects directly the target population size ξ_{it} and their trajectory over time. Table 2 presents the information

criteria (IC)—AIC, BIC and QAIC—for eight specifications of α_{it}, given yearly β_t and constant γ, based on all the data 2019–2024. See Supplementary Materials Section D for the full results related to the specification of all the parameters, some of which may be referred to below.

Table 2: AIC, BIC and QAIC by community anchor parameter α_{it}, given yearly β_t and constant γ (S1 is the time-invariant baseline with constant β as well)

		Poisson PMLE				NB-MLE	
Specification	Label	AIC	BIC	QAIC	$\hat{c}$	AIC	BIC
S1	Intercept only	22,406	22,421	2,485	18.5	5,329	5,350
S2	Year	16,471	16,539	1,846	14.1	5,206	5,279
S3	Year + sex	16,050	16,123	1,802	12.7	5,197	5,275
S4	Year x UKR	11,198	11,297	1,274	9.4	5,195	5,299
S5	Year x simplified	13,961	14,060	1,579	14.5	5,206	5,310
S6	Year + continent	12,628	12,722	1,430	10.2	5,176	5,275
S7	Year + sex + UKR	13,876	13,955	1,563	11.5	5,195	5,278
S8	Year x UKR + sex	11,176	11,280	1,273	9.0	5,186	5,295

Notes: UKR – a binary indicator equal to 1 for Ukrainian, and 0 for non-Ukrainian; Simplified – a binary indicator equal to 1 for six post-Soviet origin countries with a simplified access to Polish labour market (Ukraine, Belarus, Moldova, Georgia, Armenia, Russia; see Table A.2 in Supplementary Materials). $\hat{c}$ is the Pearson dispersion of each Poisson PMLE fit; QAIC $= -2\ell/\hat{c}_{S8} + 2(K+1)$, with K the number of parameters, uses the dispersion of the most general specification, S8 ($\hat{c}_{S8} = 9.0$) as suggested by Burnham and Anderson (2002, p. 69)

Although NB-MLE with offset has been rejected as the primary estimator for this application, their results are included here to illustrate that formal model-selection criteria such as AIC and BIC do not handle random-effects models in the same way as fixed-effects models. In particular, the ability to accommodate over-dispersion may unduly favor random-effects models. A related problem is the much narrower range of AIC (or BIC) values as the specification of α_{it} varies, which shows that less information about the mean structure is captured by these criteria.

When it comes to Poisson PMLE with offset, which is our focus here, there is a large drop of IC from yearly α (as in exploratory estimation above) to our preferred S8 specification. On the one hand, the interaction between year and Ukraine (separated from the other countries of origin) is the most important reason, as can be seen by comparing S8 to S4 specification. On the other hand, as described in Section 4, sex is a factor by which all the counts vary; the small change of IC from S4 to S8 is simply due to the overwhelming mass of males in these data.

The year-by-Ukrainian interaction in the community anchor is motivated by the 2022 EU Temporary Protection Directive, which expanded the Regularized Ukrainian population N without a similar effect on the unauthorized counts m. Without this interaction, the community anchor would have to absorb this structural break as a declining trend that affects the non-Ukrainians as well.



Figure 5: Estimated trajectory of the community anchor (α) by sex and origin, Poisson PMLE with the S8 specification. Lines are bootstrap medians; shaded bands are 80% (dark) and 95% (light) percentile CIs

In addition to the S8 specification of α_{it}, with yearly β_t and constant γ, we have also considered other specifications of β and γ. Regarding the exposure elasticity β_{it}, adding sex or Ukrainian origin to yearly β_t does not improve IC and introduces instability, whereas holding β constant across years would produce nearly identical population size trajectories by Poisson PMLE and NB-MLE, and misses the structural break noted above. Regarding the offset γ, allowing it to vary by covariates or time has a larger effect on Poisson PMLE than NB-MLE, in the sense that IC differs more for the former than the latter, which is similar to the contrast between them in Table 2, although the resulting population size estimates still fall within the confidence intervals by the S8 specification. In short, the S8 specification here balances responsiveness to structural break with

estimation stability and model parsimony.

Figure 5 displays the trajectory of the estimated community anchor (α) based on Poisson PMLE with the S8 specification. See Supplementary Materials Section D for the all discussed models, including those by NB-MLE with offset.

All the estimated α_{it} lie below unity, without any need of constraining. The estimated α-value for non-Ukrainians is stable at about 0.83–0.85 throughout 2019–2024, while that for Ukrainians tracks a similar level until 2021 and then declines sharply to about 0.67 by 2024 — consistent with the 2022 Temporary Protection Directive that greatly reduces the extent of unauthorized Ukrainians. The sex effect shifts the trajectories downward for females by a constant (about 0.03), reflecting a lower prevalence of unauthorized female migrants, which is also as anticipated.

5.3 Unauthorized population size estimates, sensitivity analyses

Table 3 presents the estimates of the unauthorized adult (18+) non-Schengen population in Poland from 2019 to 2024, stratified by sex, whereas Figure 6 shows the estimates by year and origin (Ukrainian vs non-Ukrainian). The mean structure is the S8 specification described in Section 5.2. The point estimates are given as the medians of cluster FWB, together with the 80% and 95% confidence intervals. Poisson PMLE provides the primary estimates, while the results by NB-MLE are included for the purpose of sensitivity analysis.

Our estimates of the adult non-Schengen unauthorized population size is 94,000 (80% interval: 36,000–206,000; 95% CI: 24,000–279,000) in 2019 and 101,000 (80% interval: 46,000–177,000; 95% CI: 29,000–254,000) in 2024. The ratio to the ZUS reference population size is 15.4% in 2019 and 8.9% in 2024. The trajectory of these estimates over time, including the breakdown by either sex or origin, are as can be expected from that of the community anchor parameter α shown in Figure 5. According to our Poisson PMLE estimates the unauthorized population rose by about 7% between 2019 and 2024, while the reference population almost doubled.

As shown in the Figure 6, the estimated number of unauthorized non-Ukrainian residents rose from

about 34,000 in 2019 to about 89,000 in 2024, more than two and a half times over the period and closely matching the growth of the corresponding ZUS reference population, which increased from roughly 133,000 to 362,000.

Table 3: Estimated size of unauthorised 18+ non-Schengen population by year and sex. Observed counts (Border). Reference population (Ref. pop.) by ZUS register. Model with S8 specification. Cluster FWB median with 80% and 95% percentile CIs for Poisson PMLE or NB-MLE. All figures in thousands.

				Poisson PMLE			NB-MLE		
Year	Sex	Border	Ref. pop.	Median	80% CI	95% CI	Median	80% CI	95% CI
2019	Female	1.5	208	29	(11, 65)	(7, 91)	31	(22, 48)	(18, 58)
	Male	5.1	401	64	(25, 141)	(17, 190)	92	(62, 139)	(52, 178)
	Total	*6.6*	*609*	*94*	*(36, 206)*	*(24, 279)*	*124*	*(85, 183)*	*(70, 235)*
2020	Female	0.7	236	30	(13, 57)	(8, 76)	25	(15, 41)	(11, 53)
	Male	2.7	445	63	(28, 121)	(17, 165)	70	(42, 117)	(31, 151)
	Total	*3.4*	*682*	*92*	*(41, 178)*	*(25, 233)*	*96*	*(57, 157)*	*(41, 203)*
2021	Female	0.5	285	34	(18, 59)	(11, 78)	71	(40, 130)	(30, 187)
	Male	2.6	546	72	(39, 136)	(25, 189)	222	(122, 415)	(94, 614)
	Total	*3.1*	*831*	*106*	*(58, 192)*	*(35, 261)*	*295*	*(162, 539)*	*(126, 776)*
2022	Female	0.3	486	28	(17, 40)	(12, 48)	26	(12, 55)	(7, 83)
	Male	2.6	633	56	(36, 84)	(27, 109)	78	(33, 167)	(21, 253)
	Total	*3.0*	*1,119*	*84*	*(54, 124)*	*(40, 156)*	*105*	*(45, 221)*	*(29, 336)*
2023	Female	0.5	497	34	(20, 47)	(13, 57)	32	(18, 56)	(14, 73)
	Male	3.8	663	81	(50, 124)	(35, 167)	109	(62, 191)	(45, 248)
	Total	*4.4*	*1,160*	*115*	*(71, 170)*	*(49, 221)*	*141*	*(80, 247)*	*(59, 318)*
2024	Female	1.0	467	28	(13, 47)	(8, 59)	47	(30, 75)	(21, 102)
	Male	5.8	666	72	(33, 131)	(21, 190)	176	(106, 292)	(82, 398)
	Total	*6.7*	*1,133*	*101*	*(46, 177)*	*(29, 254)*	*223*	*(136, 370)*	*(103, 494)*

The estimated size of unauthorized Ukrainians moved in the opposite direction, falling from about 60,000 to 11,000, even though the Ukrainian reference population grew from about 476,000 to 771,000. The modest overall increase therefore conceals a pronounced change in composition, with the unauthorized population shifting away from Ukrainian nationals and towards other non-Schengen nationals. All figures reported here are estimated medians obtained from the cluster bootstrap.

In comparison, the NB-MLE is unstable and can give implausibly high estimates from time to time,

such as in 2021 (the year after Covid outbreak) and 2024. Notice that the reference population size has a steady increase from 2019 to 2022, after which it has remained stable till 2024. Meanwhile, the total of detected unauthorized persons went first down from 6,600 in 2019 to about 3,000 in 2021–2022, before it climbed back to about the same level in 2024 as in 2019. The surge of NB-MLE in 2021 and 2024 has thus no obvious justification in these data.

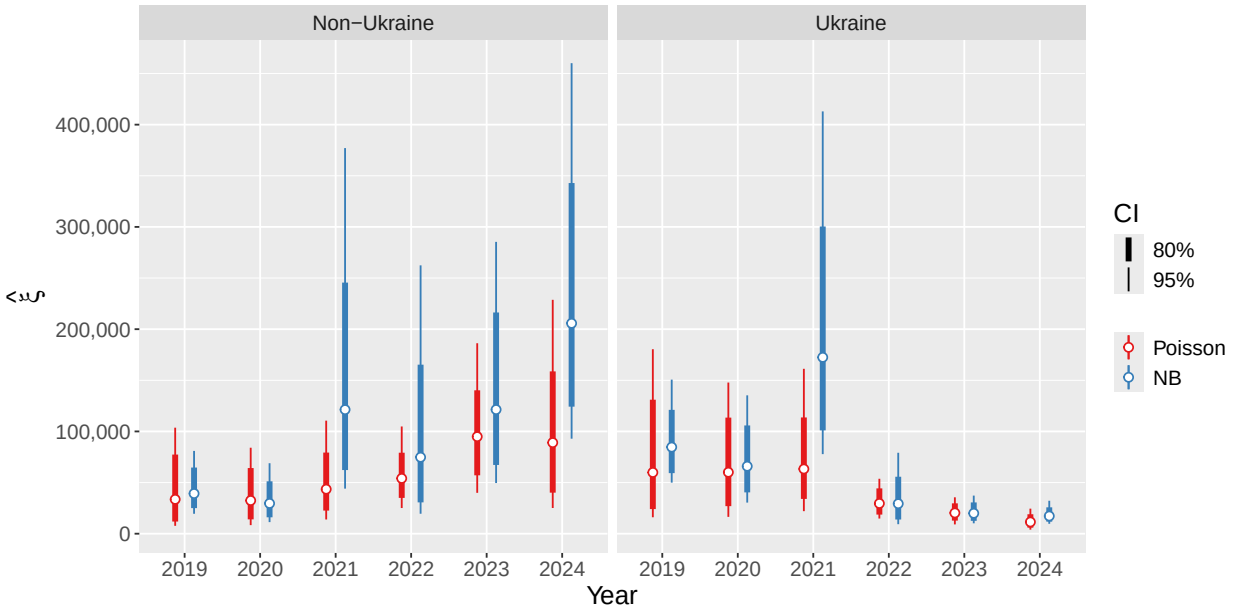


Figure 6: Estimated unauthorized population by origin (Ukrainian vs non-Ukrainian) and estimation method. Cluster bootstrap (by country) median with 80% and 95% percentile CI

Nevertheless, the ability of NB-MLE to accommodate over-dispersion in the data makes it possible to use these estimates for sensitivity analysis. In other words, had excessive over-dispersion been present and caused bias to the Poisson PMLE, how might the estimate have looked like by a method that accommodates over-dispersion? Our reported estimates are robust in this respect, since — apart from the implausible 2021 and 2024 spikes discussed above — the NB-MLE point estimates lie within the 95% confidence intervals by Poisson PMLE, whether or not one finds this necessary given the stable and plausible trajectory of Poisson PMLE.

Finally, as another plausibility check following Zhang (2008), we compare our population size

estimates with an independent administrative indicator from the Office for Foreigners that are not used in estimation. Table 4 presents the numbers of refused temporary residence decisions alongside the Poisson PMLE, which are broken down by Ukrainian vs non-Ukrainian origin. Notice that the population estimates here are the Poisson PMLE cluster-FWB medians broken down by origin, on the same basis as the sex breakdown in Table 3.

Table 4: Estimated unauthorised population size (Estimate; bootstrap median) and number of refused temporary residence decisions (Refusal), by origin (Ukrainian vs non-Ukrainian).

	Ukrainian		Non-Ukrainian	
Year	Estimate	Refusal	Estimate	Refusal
2019	59,900	19,700	33,600	13,200
2020	60,100	24,500	32,400	14,500
2021	63,300	25,400	43,500	12,300
2022	29,600	15,100	54,100	18,800
2023	20,300	2,300	94,900	21,600
2024	11,500	1,900	89,100	27,500

The most notable pattern is the sharp decline in the number of refused temporary residence decisions for Ukrainians after 2021 (from 25,400 to 15,100 in 2022 and 2,300 in 2023), reflecting the Temporary Protection Directive that regularized the status of many Ukrainian nationals. Non-Ukrainian refusals, by contrast, increased from 12,300 in 2021 to 27,500 in 2024, which is consistent with the growing non-Ukrainian population size estimate.

6 Conclusions

This paper proposes a general community-anchored modeling framework for estimating the size of unauthorized populations using only aggregated administrative data. The method requires three inputs: counts of detected unauthorized migrants, the size of an authorized foreign population (as a reference), and a covariate proxying detection intensity.

The framework is applied to the Polish data from 2019 to 2024. Our primary specification allows the community anchor (α) to vary by year, Ukrainian origin (with interaction), and sex, while

the exposure elasticity (β) varies by year only, and the offset (γ) of the detection intensity is constant throughout. In particular, our reported estimate of the adult non-Schengen unauthorized population size is 101,000 (80% interval: 46,000–177,000; 95% CI: 29,000–254,000) in 2024, which stands at 8.9% compared with the ZUS reference population. Males constitute the majority of the unauthorized population. The reported estimates are robust against potential model mis-specification, not least when one takes into account their associated confidence intervals.

The 2022 EU Temporary Protection Directive massively expanded Ukraine's reference population without a proportional increase in unauthorized Ukrainians. Our model captures this structural break naturally in terms of the year-by-Ukrainian interaction in the mean specification. This results in plausible as well as stable trajectory of the target population size over the entire period from 2019 to 2024. The framework is applicable to other large-scale regularization or protection programs that selectively inflates the lawful reference populations for specific communities.

Several limitations should be noted for practitioners. First, the wide confidence intervals reflect the inherent uncertainty in estimating hidden or hard-to-reach populations. Second, the method assumes a structural relationship between authorized and unauthorized populations that holds across a number of communities, in order for the model to be identifiable. Abrupt policy changes (such as the Temporary Protection Directive) can violate this assumption, which would require careful effort to revise any existing model. Third, the approach estimates the *stock* of unauthorized individuals at a given time, not *gross flows*; year-to-year changes in the stock reflect the net balance of entries, exits, regularizations, and removals. Fourth, the restriction to non-Schengen nationals excludes EU citizens who may overstay or work without authorization.

The framework transfers easily to other countries, just as it has been from Norway in 2006 to Poland in 2024: it requires only aggregated counts that already exist in administrative systems and can be implemented without individual-level data linkage or new data collection. The accompanying R package `uncounted` provides a complete implementation, including estimation, inference, diagnostics, and sensitivity analysis tools.

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A Migration situation in Poland

A.1 Definitions

A.1.1 Polish original version

Nielegalny pobyt – pobyt niezgodny z przepisami dotyczącymi warunków wjazdu cudzoziemców na terytorium Rzeczypospolitej Polski i ich pobytu na tym terytorium.

Nielegalny pobyt na terytorium RP ma miejsce w szczególności w sytuacji, gdy cudzoziemiec:

- Nie posiada ważnej wizy lub innego ważnego dokumentu uprawniającego go do wjazdu na terytorium Rzeczypospolitej Polskiej i pobytu na nim.
- Nie opuścił terytorium Rzeczypospolitej Polskiej po wykorzystaniu dopuszczalnego okresu pobytu.
- Nielegalnie przekroczył lub usiłował nielegalnie przekroczyć granicę.
- Nielegalnie wykonuje lub wykonywał pracę.
- Podjął działalność gospodarczą niezgodnie z obowiązującymi przepisami.
- Nie posiada wystarczających środków finansowych do pobytu na terytorium Rzeczypospolitej Polskiej.
- Został zastrzeżony do celów odmowy wjazdu w SIS lub krajowym wykazie.

Głównym skutkiem prawnym nielegalnego pobytu jest wszczęcie postępowania administracyjnego w celu **zobowiązania cudzoziemca do powrotu** (deportacji). Stosuje się następujące środki:

- **Decyzje administracyjne:**
  - **Decyzja o powrocie:** Nakłada na cudzoziemca obowiązek opuszczenia Polski. Zazwyczaj wyznacza się termin dobrowolnego wyjazdu (od 15 do 30 dni).
  - **Zakaz wjazdu:** Decyzji o powrocie towarzyszy zakaz ponownego wjazdu na terytorium RP oraz strefy Schengen na okres od 6 miesięcy do 5 lat.
  - **Wpis do SIS:** Dane są wprowadzane do Systemu Informacyjnego Schengen, co skutkuje odmową wjazdu do większości krajów europejskich.
- **Zatrzymanie i środki przymusu:**
  - **Strzeżone ośrodki:** Sąd może nakazać umieszczenie cudzoziemca w strzeżonym ośrodku dla cudzoziemców w celu zabezpieczenia wykonania decyzji o powrocie.
  - **Środki alternatywne:** Obowiązek zgłaszania się do organów Straży Granicznej, przekazanie dokumentu podróży do depozytu lub zabezpieczenie finansowe.
- **Odpowiedzialność finansowa i karna:**
  - **Grzywna:** Nielegalny pobyt stanowi wykroczenie podlegające karze grzywny.
  - **Kara pozbawienia wolności:** Nielegalne przekroczenie granicy przy użyciu siły lub posługiwanie się fałszywymi dokumentami może skutkować odpowiedzialnością karną (od 3 do 8 lat więzienia).
- **Nieważność dokumentów:** Z dniem, w którym decyzja o powrocie staje się ostateczna, posiadane wizy lub zezwolenia na pracę wygasają z mocy prawa.

A.1.2 English translation

Illegal stay (Poland) – stay not in accordance with the regulations concerning the conditions of entry of foreigners into the territory of the Republic of Poland and their stay in that territory.

Illegal stay on the territory of the Republic of Poland occurs in particular when a foreigner:

- Does not have a valid visa or other valid document entitling him/her to enter and stay on the territory of the Republic of Poland.
- Has not left the territory of the Republic of Poland after the expiry of the permitted period of stay.
- Has illegally crossed or attempted to illegally cross the border.
- Is or has been working illegally.
- Has undertaken economic activity in violation of the applicable regulations.
- Does not have sufficient financial means to stay in the territory of the Republic of Poland.
- Has been flagged for refusal of entry in the SIS or national list.

The primary legal consequence of an illegal stay is the initiation of administrative proceedings to **oblige the foreigner to return** (deportation). The following measures are applied:

- **Administrative Decisions:**
  - **Order to Return:** A decision is issued requiring the foreigner to leave Poland. A period for voluntary departure (usually 15–30 days) is typically granted.
  - **Entry Ban:** Return decisions usually include a ban on re-entering Poland and the Schengen Area for a period of 6 months to 5 years.
  - **SIS Entry:** Data is recorded in the Schengen Information System, effectively blocking entry to most European countries.
- **Detention and Restrictive Measures:**
  - **Guarded Centres:** A court may order placement in a guarded centre for foreigners (detention) to ensure deportation.
  - **Alternative Measures:** Requirement to report regularly to authorities, surrender of travel documents, or financial deposits.
- **Financial and Criminal Liability:**
  - **Fines:** Illegal stay is a misdemeanour punishable by a fine (*grzywna*).
  - **Imprisonment:** Crossing the border by force or using forged documents can lead to criminal prosecution and imprisonment (up to 3–8 years depending on the offence).
- **Legal Invalidity:** Any existing visas or work permits are automatically invalidated the moment the return decision becomes final.

A.2 Changes in the law regarding stay and work

Table S5: Key legislative changes concerning the legality and duration of stay of foreigners in Poland (2020–2024).

Year	Legal act	Document type	Extension mechanism	Group
2020	Act of 2 March 2020 (Dz.U. 2020 poz. 374, as amended)	National visas (D)	Automatic extension ex lege until 30 July 2023	All third-country nationals
2020	Same	Schengen visas issued by Poland	Legal stay extended on Polish territory until 30 July 2023 (no Schengen mobility)	All third-country nationals
2020	Same	Visa-free stay (90/180 rule)	Period of stay considered legal until 30 July 2023	All third-country nationals
2020	Same	Temporary residence permits	Validity extended ex lege until 30 July 2023	All third-country nationals
2020	Same	Residence cards (temporary, permanent, EU long-term)	Expiry tolerated; stay remained legal despite formal expiry	All third-country nationals
2020	Same	Deadlines for residence permit applications	Administrative deadlines suspended during epidemic period	All third-country nationals
2020	Same	Return decisions and obligation to leave	Suspension of deadlines for voluntary departure and enforcement	All third-country nationals
2022	Act of 12 March 2022 (Dz.U. 2022 poz. 583)	Special protection status (UKR)	Legal stay for 18 months from 24 February 2022	Ukrainian citizens
2022	EU temporary protection (Dz.U. 2022 poz. 1383 et seq.)	Temporary protection extension	Protection extended until 4 March 2024	Ukrainian citizens
2024	EU Council Decision (Dz.U. 2024 poz. 232; 854)	Temporary protection extension	Protection extended until 4 March 2025	Ukrainian citizens
2023	Amendment to Special Act on Ukraine (Dz.U. 2023 poz. 185)	Access to temp. residence after UKR stay	Temp. residence after 9 months of UKR status	Ukrainian citizens

Note:
COVID-related extensions applied ex lege and were valid until 30 days after the end of the epidemic threat state in Poland (1 July 2023). These measures regularised stay only on Polish territory and did not restore Schengen mobility rights.

Table S6: Key legislative changes concerning access to employment and duration of work authorisation for foreigners in Poland (2020–2024).

Year	Legal act	Work authorisation type	Extension / new duration	Group
2020	Act of 2 March 2020 (Dz.U. 2020 poz. 374, as amended)	Work permits (types A–E)	Validity extended until 30 July 2023	Third-country nationals
2020	Same	Employer declaration (oswiadczenie)	Continued work allowed despite expiry during epidemic	Citizens of UA, BY, MD, GE, AM, RU
2020	Same	Seasonal work permits	Automatic extension until 30 July 2023	Selected third-country nationals
2022	Act of 17 Dec. 2021 (Dz.U. 2022 poz. 91)	Employer declaration system	Max duration extended from 6 to 24 months	Citizens of UA, BY, MD, GE, AM, RU
2022	Special Act on Ukraine (Dz.U. 2022 poz. 583)	Access to labour market without work permit	Unrestricted access upon registration by employer	Ukrainian citizens
2022	Amendments to Foreigners Act	Single permit (residence and work)	Statutory 60-day deadline for completeness verification	Third-country nationals
2020–2023	COVID-19 legislation (Act of 2 March 2020 et seq.)	Modification of employment conditions	Temporary permission to change conditions without amending permits	Third-country nationals

Note:
COVID-related provisions preserved the legal validity of existing work authorisations but did not create new permits. Extensions applied until 30 days after the end of the epidemic threat state in Poland (30 July 2023).

A.3 Police records and an unauthorised stay

The Polish Police maintain the National Police Information System (pol. *Krajowy System Informacyjny Policji*, KSIP), which includes criminal records, person searches, process registrations, and traffic offenses.

In our study we use the first and third, i.e. criminal records (pol. *notowania kryminalne*) and process registrations (pol. *rejestracja procesowa*), since both concern crime: process registration records a suspect or accused person in an offense prosecuted *ex officio*, entered during pre-trial proceedings, while a criminal record links a person to a crime gathered during police work.

Unauthorized stay is not such an offense but a petty offense under Article 465 of the Act on Foreigners, adjudicated under the Code of Procedure in Cases of Petty Offenses and handled by the Border Guard. KSIP therefore records irregular migrants only when they were also linked to a crime.

It should be noted that the two registers are separate, but the services cooperate. Under the 2004 agreement between the Commanders-in-Chief of the Police and the Border Guard[8] they exchange information and share databases, and under the Act on Foreigners a foreigner detained by the Police is transferred to the Border Guard. Police contact can therefore lead to detection, which is why the police count is a natural proxy for institutional exposure, while the two counts remain registrations of different events.

[8] See legal basis here https://sip.lex.pl/akty-prawne/dzienniki-resortowe/wykazy-obowiazujacych-aktow-prawnych-komendanta-glownego-strazy-34506321.

B Border Guard and Police staffing and inspections statistics

Table S7 reports annual staffing stocks and enforcement outcomes for the Polish Border Guard and Police.

Table S7: Border Guard and Police staffing and enforcement inspection outcomes, 2019–2025

	2019	2020	2021	2022	2023	2024	2025
Police Staffing							
Total employment	98,820	97,899	100,557	101,000	96,698	97,256	102,288
Recruitments	4,655	4,373	6,684	5,169	5,174	6,411	8,509
Separations	4,567	5,234	4,053	4,726	9,457	–	–
Border Guards Staffing							
Total employment	14,718	14,496	14,845	15,098	14,858	15,488	16,557
Recruitments	409	426	767	869	1,098	1,389	1,449
Separations	544	648	418	616	1,338	759	380
Employment legality verification							
Inspections conducted [A]	4,028	2,388	2,183	1,943	1,843	1,826	1,652
Foreigners verified [B]	124,979	60,811	87,295	72,689	73,291	76,995	64,520
Illegally employed foreigners [C]	14,740	9,899	12,264	11,407	9,968	10,755	13,231
Foreigners per inspection [B/A][a]	31.0	25.5	40.0	37.4	39.8	42.2	38.1
Illegal employment detection rate [C/B][b]	11.8	16.3	14.0	15.7	13.6	14.0	20.1
Legality of stay verification							
Inspections conducted [D]	26,757	17,691	15,686	16,267	20,988	26,162	30,445
Foreigners verified [E]	61,355	40,536	39,714	43,549	57,929	68,229	81,239
Illegally staying foreigners [F]	11,153	5,823	9,490	6,984	11,152	13,935	13,377
Foreigners per inspection [E/D][a]	2.3	2.3	2.5	2.7	2.8	2.6	2.7
Illegally staying detection rate [F/E][b]	18.2	14.4	23.9	16.0	19.3	20.4	16.5

[a] Average number of foreigners verified per inspection;

[b] Share of illegally working/present foreigners among all verified, in %.

The table reports annual staffing stocks and enforcement outcomes for the Polish Border Guard. Total employment refers to the headcount at year-end. Recruitments and separations refer to officers joining and leaving the service during the calendar year. Inspections conducted refers to the number of employer or entity premises visited. Foreigners verified refers to the total number of individual foreigners whose status was checked during those inspections. Illegally employed (staying) foreigners refers to those found to be in violation of applicable regulations at the time of inspection. Population of verified migrants staying and working illegally are disjoined with an overlap around 9% (for 2025). Dashes (–) indicate data not available or not collected under this reporting category in the given year. These figures may contain duplicates and therefore slightly differ from those presented in the main text.

C Model assumptions and details

C.1 Restriction of the population

The following countries are excluded from the model:

- Croatia, Bulgaria and Romania (EU members, outside Schengen for all or part of 2019–2024)
- Ireland and Cyprus (EU members outside Schengen);
- United Kingdom: excluded for 2019–2020, retained from 2021.

Every excluded country has zero m and a positive N.

C.2 Zero-count prevalence

Table S8 presents verifications of assumptions on whether reference, lawful population is zero while unauthorised is present.

Table S8: Country-year cells with a zero reference count (N = 0) despite observed apprehensions (m > 0), for 18+ non-Schengen countries.

Year	Number of Countries	Total apprehensions (Sum of m)	Sum n (Police)
PESEL			
2023	1	1	0
Tax			
2019	2	35	12
2020	1	1	2
2023	2	2	1
ZUS			
2019	10	20	10
2020	5	7	3
2021	3	4	1
2022	4	12	23
2023	4	9	31
2024	8	26	30

Note: m = border apprehensions; n = police records; N = reference register count. Missing countries – PESEL register: Only Hong Kong (2023) is absent.

The Tax register omits very few origins: only Nepal (2019, with 34 apprehensions), Kosovo (one apprehension in each of three years) and Hong Kong (2023). The ZUS gaps are broader and cluster regionally, in particular

- Sub-Saharan African origins such as DR Congo (apprehended in five of the six years, 20 in total), Guinea-Bissau, Eritrea, Liberia, Benin and Equatorial Guinea,
- Western Balkan origins, Kosovo (five years), Serbia and Bosnia & Herzegovina. Serbia is the clearest under-coverage rather than true absence, no ZUS reference, yet 19 apprehensions and

70 police records over 2022–2024 — indicating a population that is present and policed but missing from the social-insurance register,

- Isolated, single cases: Paraguay (9 in 2019), UAE, Bahrain, St Vincent & Grenadines, Dominica, Trinidad & Tobago, Hong Kong.

Some of these discrepancies may also reflect differences in how citizenship is coded across sources rather than genuine absence from a register: the Border Guard, police and reference data may use different country-coding schemes, so an origin recorded explicitly in the apprehension data can be folded into a residual category in a register and appear as a zero. This matters most for ZUS, which assigned a sizeable share of insured foreigners to an "other/unknown" citizenship category, mechanically depressing the counts for individually named countries and inflating the apparent number of zero-reference cells.

Table S9 reports the share of country-of-origin groups with zero counts in apprehension (m) and police (n) records, by sex and year. Females have higher zero rates than males: only 17–31% of countries have positive counts in both sources for females, compared to 50–65% for males. This sparsity motivates the γ offset, which allows cells with $n_{it} = 0$ to enter the likelihood without the regrouping required by Zhang (2008).

Table S9: Zero counts in apprehension and police records by sex and year for 18+ non-Schengen countries.

Year	Sex	m=0 (%)	m=0 (C)	n=0 (%)	n=0 (C)	Both>0 (%)	Both>0 (C)	Countries
2019	Males	27.6	29	27.6	29	64.8	68	105
	Females	43.5	40	65.2	60	27.2	25	92
	Total	*27.5*	*30*	*28.4*	*31*	*66.1*	*72*	*109*
2020	Males	37.8	42	34.2	38	52.3	58	111
	Females	61.7	58	73.4	69	19.1	18	94
	Total	*36.6*	*41*	*32.1*	*36*	*53.6*	*60*	*112*
2021	Males	48.7	57	27.4	32	49.6	58	117
	Females	66.3	65	71.4	70	17.3	17	98
	Total	*48.3*	*57*	*28.0*	*33*	*50.0*	*59*	*118*
2022	Males	45.2	57	31.7	40	52.4	66	126
	Females	70.8	85	73.3	88	17.5	21	120
	Total	*43.7*	*55*	*30.2*	*38*	*54.0*	*68*	*126*
2023	Males	29.9	38	26.0	33	64.6	82	127
	Females	61.3	76	67.7	84	24.2	30	124
	Total	*26.8*	*34*	*25.2*	*32*	*66.1*	*84*	*127*
2024	Males	29.6	37	31.2	39	61.6	77	125
	Females	51.3	59	63.5	73	31.3	36	115
	Total	*32.1*	*42*	*32.8*	*43*	*59.5*	*78*	*131*

Note: m denotes border apprehensions; n denotes police records. Reference population: ZUS. C = number of countries. The last column denotes the number of countries with a positive ZUS count for a given sex in a given year. Total rows aggregate m and n over sex within a country, so a country counts as $m = 0$ ($n = 0$) only if it has no apprehensions (police records) of either sex; the last column is then the number of countries with a positive ZUS count for at least one sex. Country includes also categories such as stateless or unknown.

C.3 Assumptions: alternative reference populations and breakdowns

C.3.1 Year and sex – PESEL and Tax registers

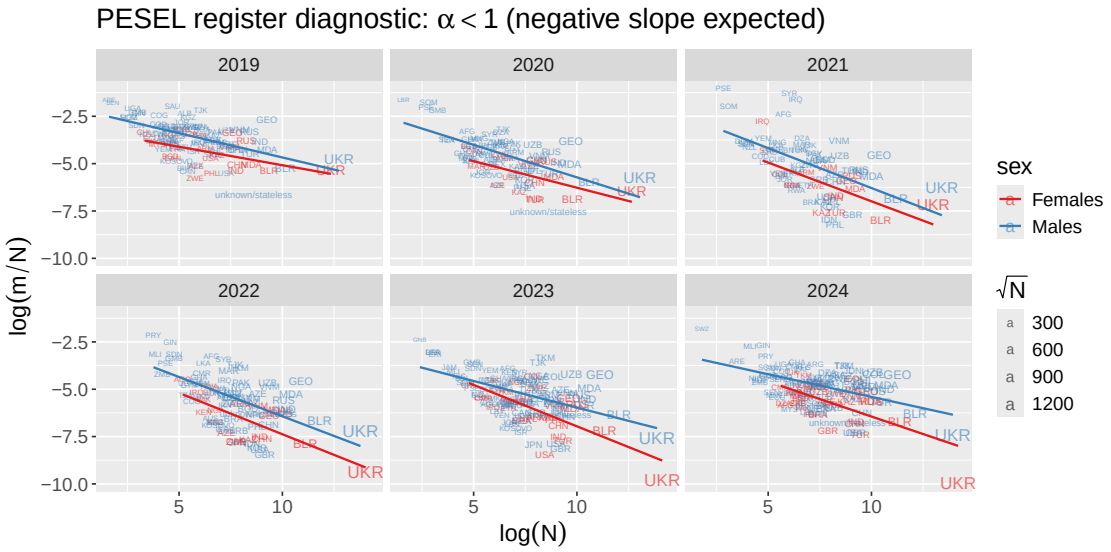


Figure S7: Diagnostic plot for community anchor ($\hat{\alpha}$): $\log(\text{Border/PESEL})$ vs $\log(\text{PESEL})$, by year and sex. Point size proportional to reference population. Slopes estimated from (6) for each panel separately.

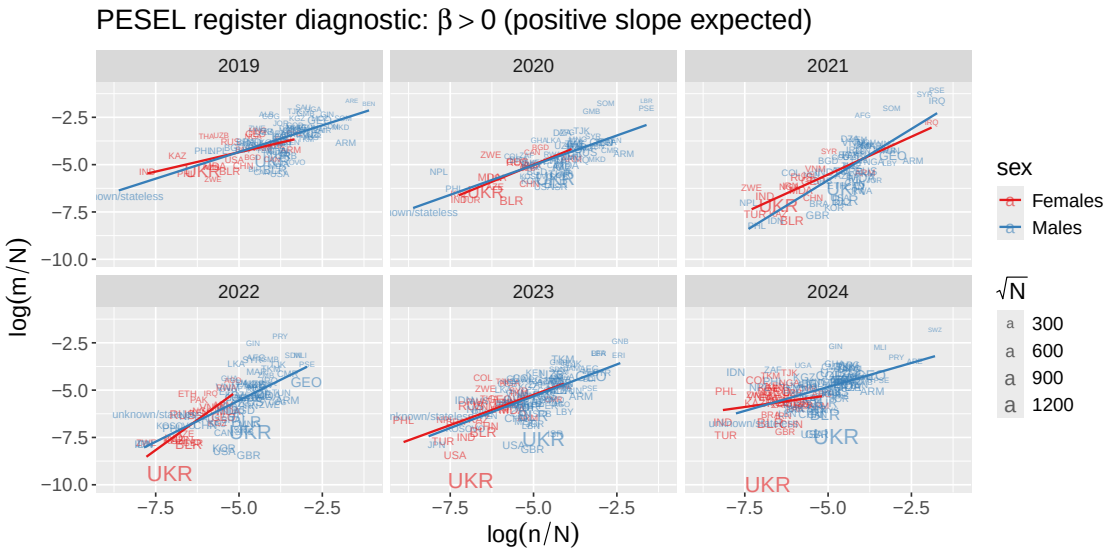


Figure S8: Diagnostic plot for exposure elasticity ($\hat{\beta}$): $\log(\text{Border/PESEL})$ vs $\log(\text{Police/PESEL})$, by year and sex. Point size proportional to reference population. Slopes estimated from (6) for each panel separately.

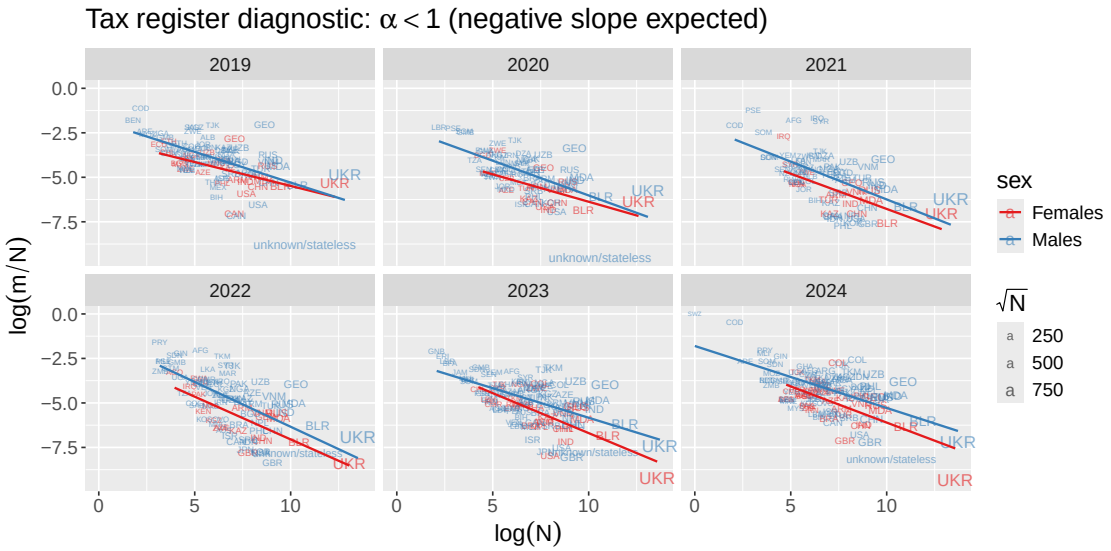


Figure S9: Diagnostic plot for community anchor ($\hat{\alpha}$): $\log(\text{Border/Tax})$ vs $\log(\text{Tax})$, by year and sex. Point size proportional to reference population. Slopes estimated from (6) for each panel separately.

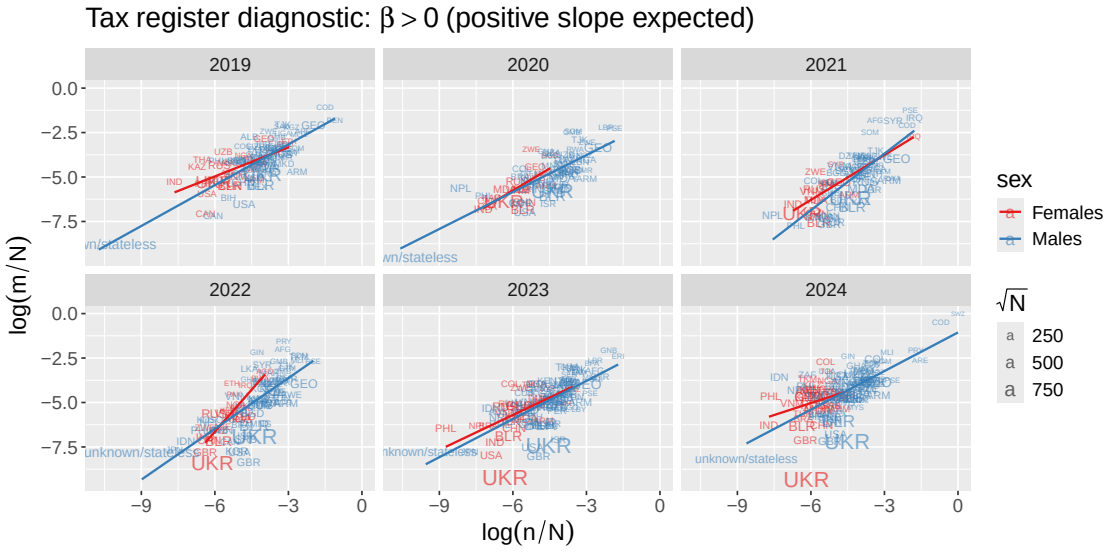


Figure S10: Diagnostic plot for exposure elasticity ($\hat{\beta}$): $\log(\text{Border/Tax})$ vs $\log(\text{Police/Tax})$, by year and sex. Point size proportional to reference population. Slopes estimated from (6) for each panel separately.

C.3.2 Year and continent (all registers)

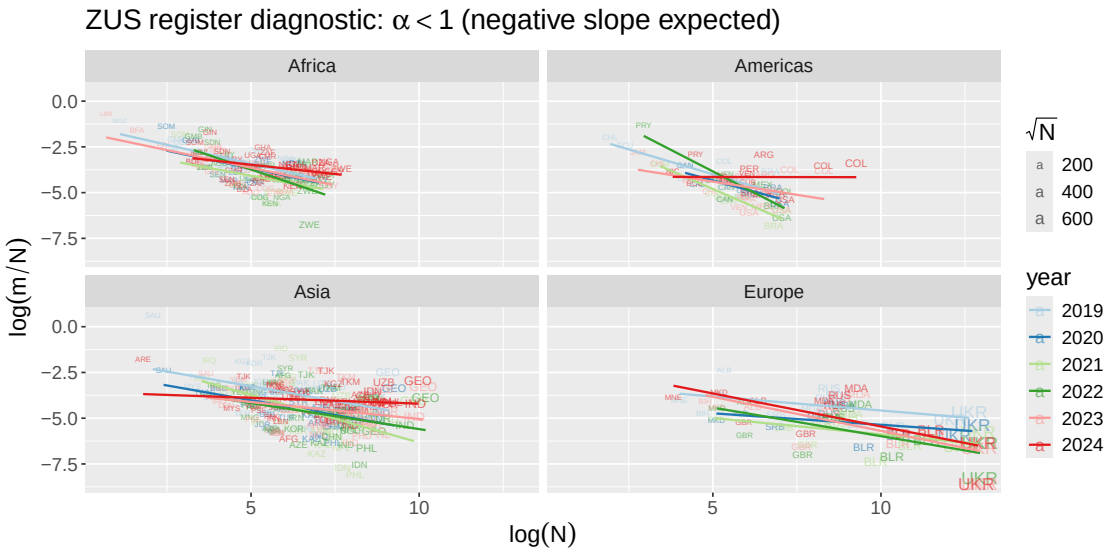


Figure S11: Diagnostic plot for community anchor ($\hat{\alpha}$): $\log(\text{Border/ZUS})$ vs $\log(\text{ZUS})$, by year and continent. Slopes estimated from (6) for each panel separately. Oceania and stateless/unknown are suppressed due to small sample sizes.

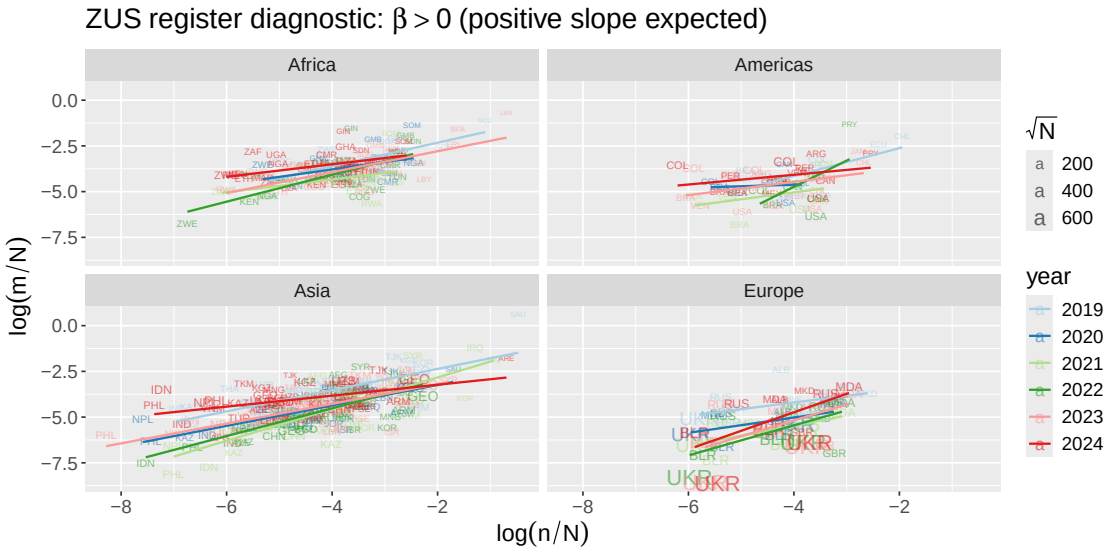


Figure S12: Diagnostic plot for exposure elasticity ($\hat{\beta}$): $\log(\text{Border/ZUS})$ vs $\log(\text{Police/ZUS})$, by year and continent. Slopes estimated from (6) for each panel separately. Grouping as in Figure S11.

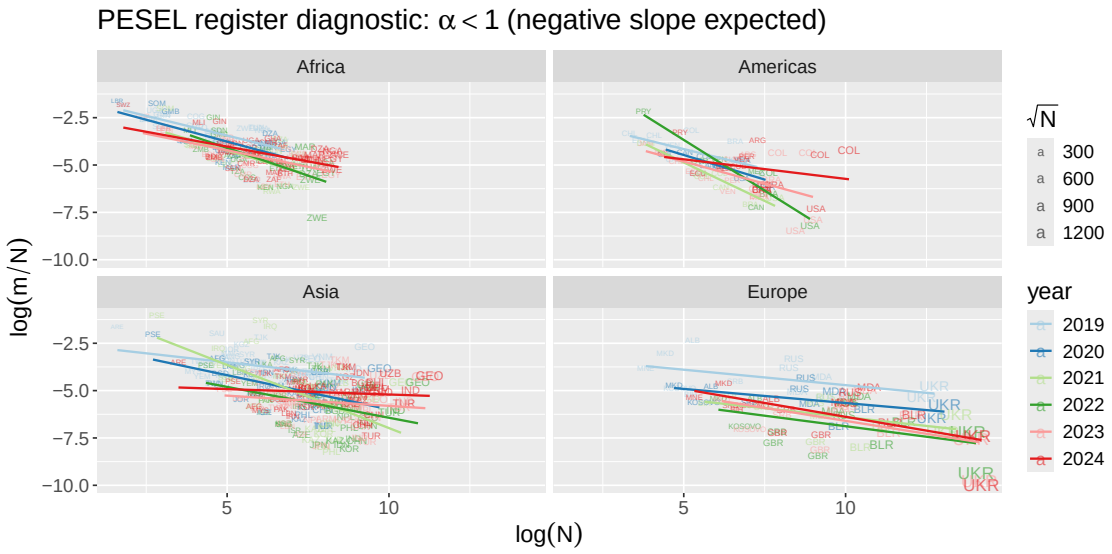


Figure S13: Diagnostic plot for community anchor ($\hat{\alpha}$): $\log(\text{Border/PESEL})$ vs $\log(\text{PESEL})$, by year and continent. Slopes estimated from (6) for each panel separately. Grouping as in Figure S11.

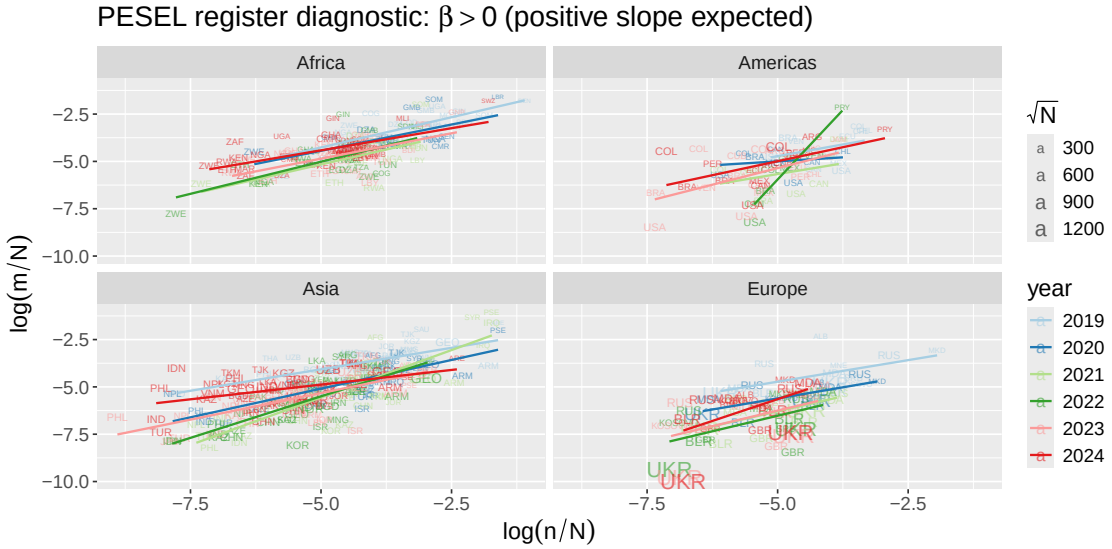


Figure S14: Diagnostic plot for exposure elasticity ($\hat{\beta}$): $\log(\text{Border/PESEL})$ vs $\log(\text{Police/PESEL})$, by year and continent. Slopes estimated from (6) for each panel separately. Grouping as in Figure S11.

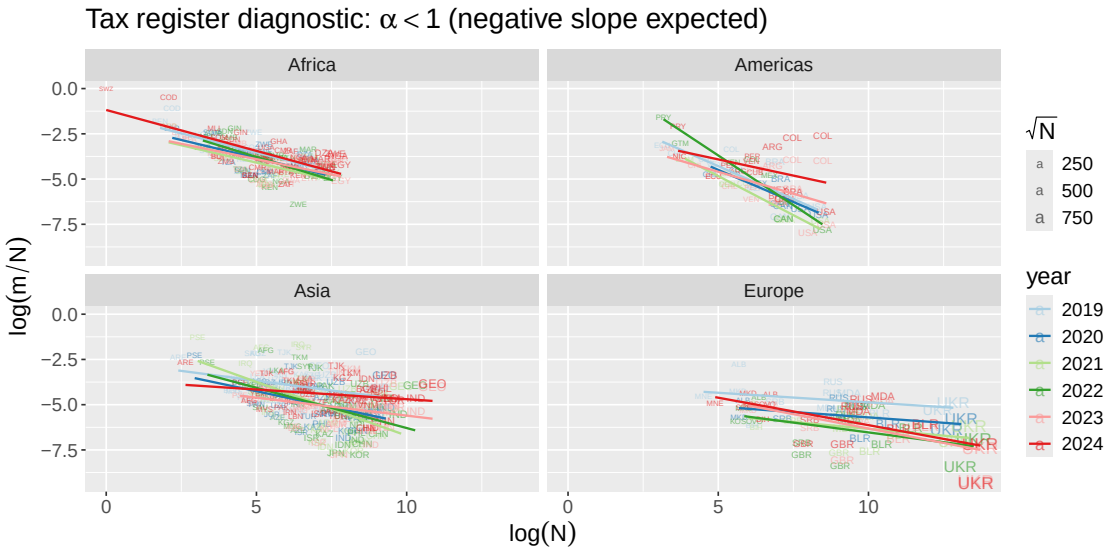


Figure S15: Diagnostic plot for community anchor ($\hat{\alpha}$): $\log(\text{Border/Tax})$ vs $\log(\text{Tax})$, by year and continent. Slopes estimated from (6) for each panel separately. Grouping as in Figure S11.

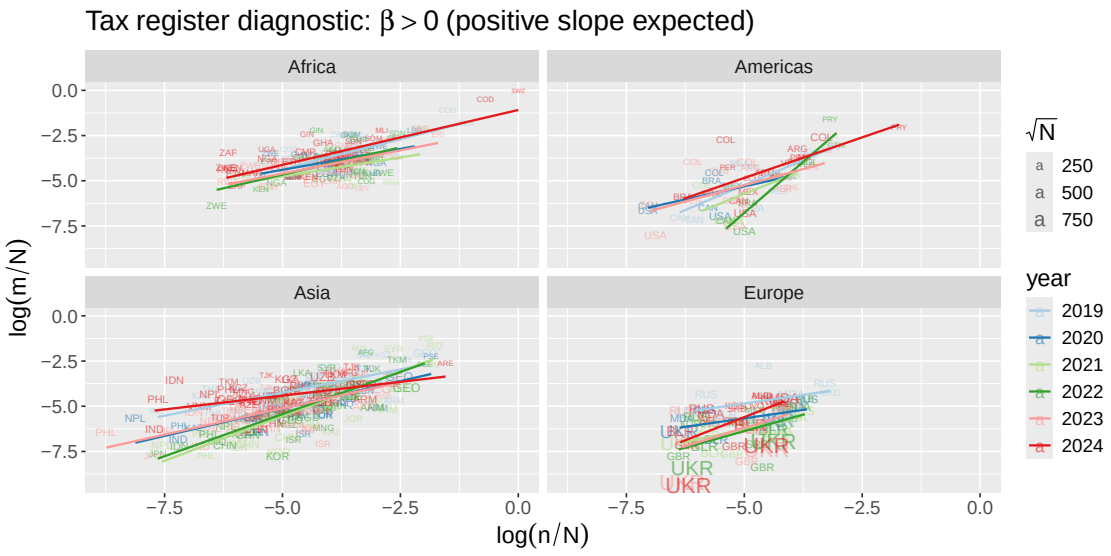


Figure S16: Diagnostic plot for exposure elasticity ($\hat{\beta}$): $\log(\text{Border/Tax})$ vs $\log(\text{Police/Tax})$, by year and continent. Slopes estimated from (6) for each panel separately. Grouping as in Figure S11.

D Details about the models

D.1 Why do we prefer the Poisson model over the Negative Binomial model?

Below we present several reasons for preferring the Poisson PMLE to NB-MLE in practice, specially when non-trivial offset $\gamma > 0$ is allowed in (5).

Offset-Gamma confounding. A manifestation of over-dispersion in the present context is the data points with $m_{it} > 0$ but $n_{it} = 0$. This can be partly dealt with by the non-trivial offset γ, and partly by the random effects u_{it} in the Poisson-Gamma mixture. This confounding would affect the estimated mean structure through γ. As we document in Supplementary Materials Section E.5, the profile log-likelihood of γ can be rather flat for the NB-MLE, whilst it peaks sharply for the Poisson PMLE in comparison. More on this in the simulation study in Section F.1.

Small-counts sensitivity. The ability of NB-MLE to accommodate over-dispersion carries a drawback as it becomes sensitive to the effects of small counts m_{it}. See the relevant measures in Supplementary Materials Section E, and the consequences on population size estimation. When NB-MLE is applied with t-specific parameter (4), it can lead to implausible spikes and disruptions of the estimated population size over time, compared to the Poisson PMLE. Notice that, since the extent of small counts is much reduced when 'rest' communities are created from the data points with $m_{it}n_{it} = 0$, this problem is somewhat related to offset-Gamma confounding above.

Robustness of PMLE. The Poisson PMLE requires only correct specification of the mean of m_{it} for consistency (Santos Silva and Tenreyro, 2006); however, see Supplementary Materials Section F and Section F.1 for finite-sample caveats. A useful indication of robust mean estimation by Poisson PMLE is whether the associated confidence interval includes the NB-MLE. Notice that, in practice, one may find the trajectory of unauthorized population size estimates to be more reliable by Poisson PMLE than NB-MLE, even when the former is not more 'correct' for each given period.

We conducted exploratory estimation separately for each year, without any parameterizations that connect the parameter values of (α, β) over time. Three estimators are used to this end: the original NB-MLE of Zhang (2008), with a different 'rest' community in each year but without offset ($\gamma \equiv 0$), the Poisson PMLE allowing for a non-trivial offset $\gamma > 0$, the NB-MLE with $\gamma > 0$ and Poisson-Gamma mixture at the same time.

Figure S17 compares the three. The associated confidence intervals are all given by (10), where the Y-axis is clipped at 450,000 for readability. Clearly, the NB-MLE with offset is too unstable to be acceptable, due to the reasons that have been explained and illustrated earlier. Separate yearly NB-MLE without offset and Poisson PMLE with offset are in better agreement with each other. However, NB-MLE without offset makes less efficient use of the positive counts m when $n = 0$, which amounts to 155 out of 1,354 data points here. Moreover, the Poisson PMLE shows a decreasing trajectory from 2022 onwards, which is plausible in light of the 2022 EU Temporary Protection Directive concerning the Ukrainian. The fact that this is achieved with 'free' parameters over time lends considerable plausibility to the mean structure of Poisson PMLE, which is untampered by data regrouping required for NB-MLE without offset.

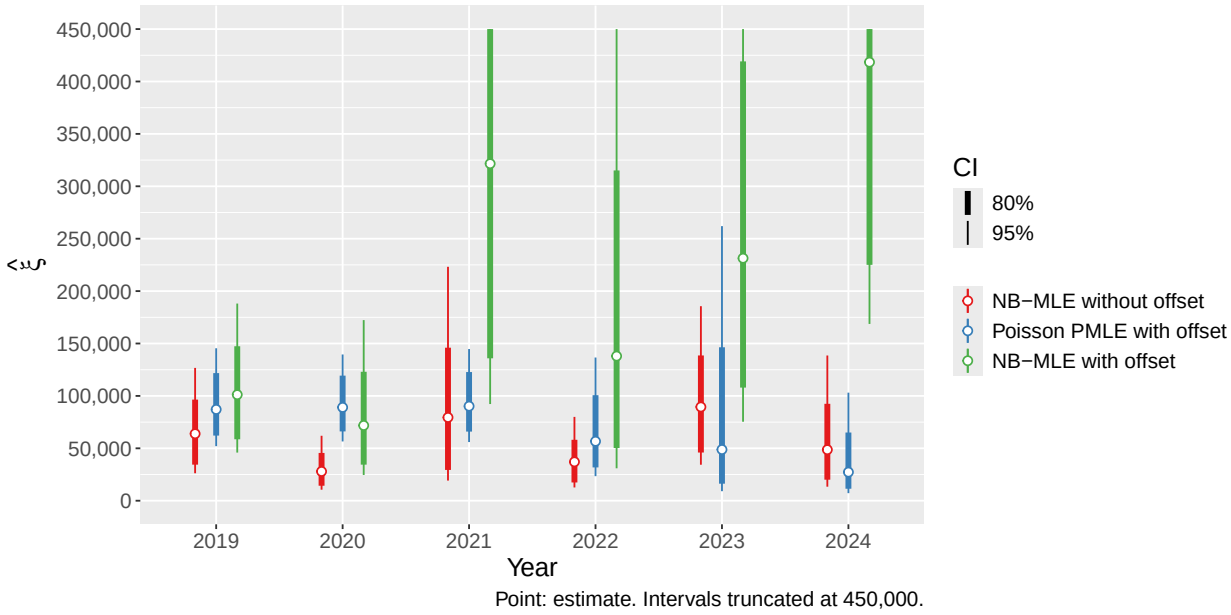


Figure S17: Separate yearly estimation of unauthorized population total by NB-MLE without offset, Poisson PMLE with offset, and NB-MLE with offset. Two confidence intervals by (10) reported

D.2 AIC/BIC across eight specifications and three registers

Table S10: AIC, BIC and QAIC for eight covariate specifications (S1–S8) across three registers and two estimation methods.

		Poisson PMLE				NB-MLE	
Spec	Label	AIC	BIC	QAIC	$\hat{c}$	AIC	BIC
ZUS							
S1	Intercept only	22,405.8	22,421.4	2,484.7	18.53	5,329.4	5,350.2
S2	Year	16,471.3	16,539.1	1,846.3	14.11	5,205.8	5,278.8
S3	Year + sex	16,050.1	16,123.1	1,801.6	12.74	5,197.2	5,275.4
S4	Year x UKR	11,198.2	11,297.2	1,274.0	9.38	5,195.1	5,299.3
S5	Year x simplified	13,960.8	14,059.8	1,579.4	14.50	5,206.2	5,310.4
S6	Year + continent	12,628.1	12,721.9	1,430.3	10.17	5,175.7	5,274.7
S7	Year + sex + UKR	13,876.5	13,954.6	1,563.0	11.55	5,194.8	5,278.2
S8	Year x UKR + sex	11,176.0	11,280.2	1,273.3	9.04	5,186.0	5,295.4
Tax							
S1	Intercept only	24,257.4	24,273.9	2,265.4	17.81	5,758.9	5,780.9
S2	Year	18,116.4	18,187.8	1,711.9	14.37	5,636.3	5,713.3
S3	Year + sex	17,992.8	18,069.8	1,702.2	13.15	5,629.9	5,712.4
S4	Year x UKR	13,060.5	13,164.9	1,252.2	10.23	5,633.2	5,743.2
S5	Year x simplified	16,034.7	16,139.2	1,529.0	14.85	5,637.7	5,747.7
S6	Year + continent	14,282.7	14,381.7	1,364.1	8.83	5,557.8	5,662.2
S7	Year + sex + UKR	16,339.1	16,421.6	1,550.1	12.50	5,631.7	5,719.7
S8	Year x UKR + sex	13,046.4	13,156.3	1,252.7	10.74	5,626.1	5,741.6
PESEL							
S1	Intercept only	24,188.6	24,205.1	3,550.6	17.08	5,746.4	5,768.4
S2	Year	17,226.2	17,297.8	2,547.7	11.65	5,569.5	5,646.6
S3	Year + sex	16,875.2	16,952.3	2,498.0	10.05	5,552.4	5,635.1
S4	Year x UKR	12,078.3	12,183.0	1,803.8	7.23	5,559.7	5,670.0
S5	Year x simplified	15,312.4	15,417.1	2,277.6	10.16	5,567.6	5,677.8
S6	Year + continent	14,114.7	14,213.9	2,100.4	8.29	5,523.5	5,628.2
S7	Year + sex + UKR	15,144.3	15,227.0	2,246.1	9.01	5,552.8	5,641.0
S8	Year x UKR + sex	12,042.0	12,152.2	1,800.2	6.83	5,541.7	5,657.4

Note: $\hat{c}$ = Pearson dispersion (chi-square over residual df) of the Poisson PMLE fit. QAIC $= -2\ell/\hat{c}_{S8} + 2(K+1)$, with K the number of parameters (including γ) and $\hat{c}_{S8}$ the dispersion of the most general specification in each register (ZUS: 9.0; Tax: 10.7; PESEL: 6.8). Not defined for NB-MLE, which models the dispersion.

Across all three registers, S8 (α varying by year $\times$ Ukrainian origin plus a sex effect) has the lowest Poisson PMLE information criteria. The NB-MLE criteria differ little across specifications, with S8 within a few units of the best NB fit in every register, so we adopt S8 for both methods.

D.3 Beta covariate selection

The primary specification (S8) allows the exposure elasticity to vary by year only. Here we examine whether extending β with sex and/or Ukrainian origin improves the model. All specifications below share the same S8 community anchor structure (α varying by year $\times$ Ukrainian origin, plus a sex effect).

Table S11: Effect of beta specification on AIC, BIC and QAIC across three registers. S8 is the primary specification. All specifications share alpha = year x UKR + sex; k = number of parameters.

Model	Spec	Covariates in beta	AIC	BIC	QAIC	$\hat{c}$	k
ZUS							
NB-MLE	S8	year	5,186.0	5,295.4			21
NB-MLE	S9	year + sex	5,186.8	5,301.5			22
NB-MLE	S10	year + UKR	5,187.9	5,302.6			22
NB-MLE	S11	year + sex + UKR	5,188.8	5,308.7			23
Poisson PMLE	S8	year	11,176.0	11,280.2	1,273.3	9.04	20
Poisson PMLE	S9	year + sex	11,169.1	11,278.6	1,274.3	8.97	21
Poisson PMLE	S10	year + UKR	11,134.7	11,244.2	1,270.5	8.95	21
Poisson PMLE	S11	year + sex + UKR	11,040.3	11,154.9	1,261.8	8.59	22
Tax							
NB-MLE	S8	year	5,626.1	5,741.6			21
NB-MLE	S9	year + sex	5,628.1	5,749.1			22
NB-MLE	S10	year + UKR	5,628.1	5,749.1			22
NB-MLE	S11	year + sex + UKR	5,630.1	5,756.6			23
Poisson PMLE	S8	year	13,046.4	13,156.3	1,252.7	10.74	20
Poisson PMLE	S9	year + sex	13,029.5	13,145.0	1,252.9	10.73	21
Poisson PMLE	S10	year + UKR	12,909.4	13,024.9	1,241.7	10.37	21
Poisson PMLE	S11	year + sex + UKR	12,677.9	12,798.9	1,222.0	10.38	22
PESEL							
NB-MLE	S8	year	5,541.7	5,657.4			21
NB-MLE	S9	year + sex	5,541.3	5,662.5			22
NB-MLE	S10	year + UKR	5,543.6	5,664.9			22
NB-MLE	S11	year + sex + UKR	5,543.2	5,670.0			23
Poisson PMLE	S8	year	12,042.0	12,152.2	1,800.2	6.83	20
Poisson PMLE	S9	year + sex	12,038.8	12,154.5	1,801.4	6.84	21
Poisson PMLE	S10	year + UKR	11,881.1	11,996.8	1,778.3	6.66	21
Poisson PMLE	S11	year + sex + UKR	11,819.4	11,940.6	1,771.0	6.61	22

Note: $\hat{c}$ = Pearson dispersion of the Poisson PMLE fit; QAIC $= -2\ell/\hat{c}_{S8} + 2(k+1)$ with $\hat{c}_{S8}$ the dispersion of the adopted specification S8 in each register (ZUS: 9.0; Tax: 10.7; PESEL: 6.8). Not defined for NB-MLE, which models the dispersion.

Extending β with sex and Ukrainian origin (S9–S11) lowers the Poisson AIC and BIC in every register, but not all QAIC and leaves the NB AIC essentially unchanged (and marginally higher). Because the additional β terms improve only the Poisson fit while S8 already minimises or nearly minimises the NB AIC, we keep β varying by year only, giving a parsimonious specification that is comparable across both methods and all three registers. Figure S18 shows the comparison to all S1-S11 specifications.

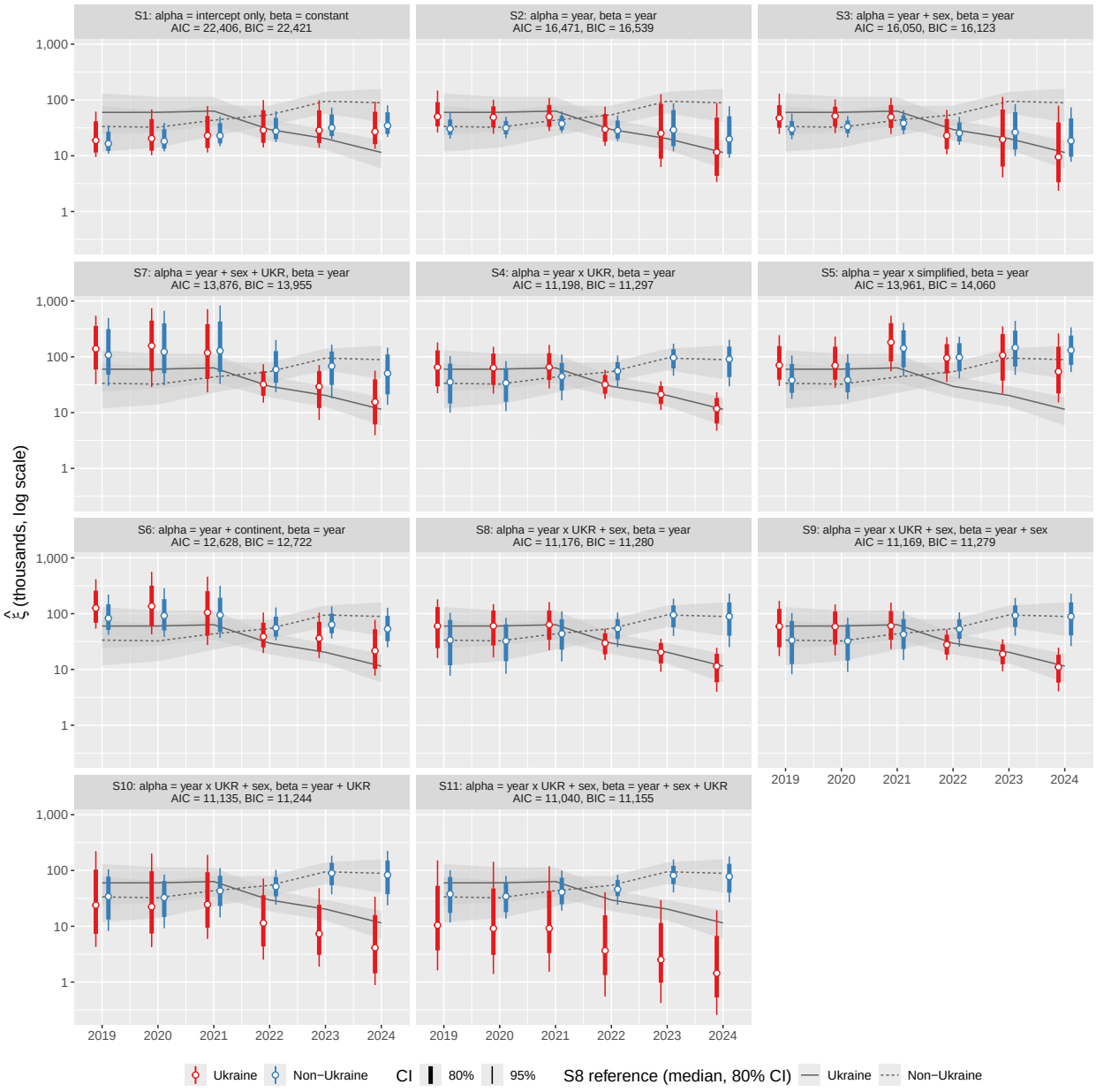


Figure S18: Estimated unauthorised population by year and origin under specifications S1–S11 (Poisson PMLE, ZUS reference population). Points are cluster fractional-weighted bootstrap medians (R = 999, clustered by country) with 80% (thick) and 95% (thin) percentile confidence intervals, on a logarithmic scale. Each panel header gives the specification of α and β and the AIC and BIC of the fit. The S8 median (line) and its 80% interval (grey band) are repeated in every panel as the reference; solid for Ukrainian and dashed for non-Ukrainian origin.

Despite S11 has the lowest AIC, BIC and QAIC among the β specifications, yet we do not adopt it. Figure S18 and Table S12 shows why.

Table S12: S8 versus S11 (Poisson PMLE, ZUS reference): fit, the Ukrainian term in the exposure elasticity, the range of the fitted Ukrainian detection rate, and cluster-FWB estimates (median with 95% percentile CI, thousands) by origin.

Spec	QAIC	$\hat{\beta}_{\text{UKR}}$ (SE)	$\hat{\rho}$, UKR cells	$\hat{\xi}$ UKR 2019	$\hat{\xi}$ UKR 2024	$\hat{\xi}$ non-UKR 2024	Failed replicates
S8	1,273.3	–	0.01–0.09	60 (16, 180)	11 (4, 24)	89 (25, 229)	5
S11	1,261.8	-0.54 (0.51)	0.23–1.26	10 (2, 152)	1 (0, 19)	78 (27, 178)	315

Note: QAIC with $\hat{c}$ from S8. SE = HC1 robust. $\hat{\rho}$ is admissible only if at most 1. Failed replicates = bootstrap draws (of 999) in which the fit did not converge. S8: $\alpha \sim$ year $\times$ UKR + sex, $\beta \sim$ year; S11: same α, $\beta \sim$ year + sex + UKR.

The coefficient β is not insignificant (-0.54, SE 0.51), but it raises the fitted Ukrainian detection rate β from at most 0.09 to between 0.23 and 1.26, exceeding one for Ukrainian women in 2024, and correspondingly reduces the Ukrainian estimate from 60 to 10 thousand in 2019 and from 11 to 1 thousand in 2024. Furthermore, 315 of 999 bootstrap replicates fail to converge, against 5 under S8, while the non-Ukrainian estimates are essentially unchanged. The improvement in fit therefore comes from a part of the model in which the decomposition of m into ξ and ρ is no longer identified.

D.4 Coefficient estimates

Table S13 presents the S8 coefficient estimates under Poisson PML and unconstrained NB. Both columns are on the original α and β scale and are directly comparable. The constrained NB is also fit (with $\alpha \in (0, 1)$ enforced via a logit link) but its coefficients are on the link scale and therefore not magnitude-comparable to the other two; we report the resulting α values per cell — already back-transformed — in Table S14, alongside the Poisson and NB values for direct comparison.

Table S13: S8 coefficient estimates under Poisson PML and unconstrained NB. Cluster fractional-weighted bootstrap medians with 95% percentile confidence intervals in parentheses (R = 999, clustered by country). ZUS reference population. Both models unconstrained.

	Poisson	NB
Community anchor ($\hat{\alpha}$)		
Intercept (2019, female, non-UKR)	0.816 (0.617, 0.959)	0.819 (0.724, 0.911)
Year 2020	-0.016 (-0.058, 0.024)	-0.050 (-0.168, 0.039)
Year 2021	-0.013 (-0.114, 0.121)	0.086 (-0.021, 0.212)
Year 2022	-0.014 (-0.128, 0.147)	0.003 (-0.168, 0.139)
Year 2023	0.035 (-0.077, 0.169)	0.038 (-0.060, 0.137)
Year 2024	0.012 (-0.073, 0.147)	0.086 (-0.001, 0.177)
Ukrainian origin	0.004 (-0.056, 0.094)	0.013 (-0.039, 0.068)
Male	0.017 (-0.010, 0.042)	0.043 (0.011, 0.074)
Year 2020 × Ukrainian	0.009 (-0.011, 0.027)	0.024 (-0.013, 0.075)
Year 2021 × Ukrainian	-0.000 (-0.069, 0.043)	-0.048 (-0.114, 0.007)
Year 2022 × Ukrainian	-0.077 (-0.142, -0.028)	-0.116 (-0.184, -0.036)
Year 2023 × Ukrainian	-0.154 (-0.214, -0.111)	-0.182 (-0.233, -0.132)
Year 2024 × Ukrainian	-0.174 (-0.231, -0.143)	-0.237 (-0.287, -0.192)
Exposure elasticity ($\hat{\beta}$)		
Intercept (2019)	0.718 (0.342, 1.155)	0.863 (0.661, 1.202)
Year 2020	0.155 (0.058, 0.275)	0.128 (-0.108, 0.323)
Year 2021	0.253 (0.026, 0.539)	0.492 (0.270, 0.764)
Year 2022	0.210 (-0.043, 0.513)	0.251 (-0.120, 0.540)
Year 2023	0.199 (-0.053, 0.473)	0.228 (0.029, 0.431)
Year 2024	-0.007 (-0.211, 0.292)	0.159 (-0.010, 0.348)
Fit statistics		
Num.Obs.	1,354	1,354
AIC	11176.0	5186.0
BIC	11280.2	5295.4
QAIC	1273.3	
Dispersion ($\hat{c}$)	9.04	
Log.Lik.	-5568.003	-2572.003

Table S14 shows the actual α values per cell (year × Ukrainian origin × sex) under all three specifications: Poisson PML, unconstrained NB, and constrained NB (back-transformed via logit^{-1}). Reporting all three on a common scale makes the constrained-vs-unconstrained NB comparison clear: the constraint barely binds on Polish data (the unconstrained NB already produces $\alpha < 1$ in every cell). The unconstrained NB nonetheless pushes α close to 1 for non-Ukrainian males in recent years (0.96 in 2024), which is the primary driver of the higher NB population size estimates documented in the main text.

Table S14: Alpha values per (year, Ukrainian origin, sex) cell under three specifications. Constrained NB back-transformed via logit-inverse.

Year	Origin	Sex	NB	NB (constrained)	Poisson
2019	UKR	Females	0.8302	0.8404	0.8239
2019	UKR	Males	0.8734	0.8880	0.8349
2019	Non-UKR	Females	0.8171	0.8319	0.8180
2019	Non-UKR	Males	0.8603	0.8817	0.8290
2020	UKR	Females	0.8021	0.8067	0.8158
2020	UKR	Males	0.8453	0.8627	0.8268
2020	Non-UKR	Females	0.7644	0.7635	0.7991
2020	Non-UKR	Males	0.8075	0.8294	0.8101
2021	UKR	Females	0.8666	0.8539	0.8071
2021	UKR	Males	0.9098	0.8980	0.8181
2021	Non-UKR	Females	0.9021	0.8796	0.7983
2021	Non-UKR	Males	0.9452	0.9167	0.8093
2022	UKR	Females	0.7080	0.6684	0.7355
2022	UKR	Males	0.7512	0.7522	0.7465
2022	Non-UKR	Females	0.8070	0.7628	0.8038
2022	Non-UKR	Males	0.8502	0.8289	0.8148
2023	UKR	Females	0.6845	0.6607	0.7062
2023	UKR	Males	0.7277	0.7458	0.7172
2023	Non-UKR	Females	0.8543	0.8594	0.8518
2023	Non-UKR	Males	0.8974	0.9020	0.8628
2024	UKR	Females	0.6781	0.6579	0.6610
2024	UKR	Males	0.7213	0.7434	0.6720
2024	Non-UKR	Females	0.9030	0.9389	0.8281
2024	Non-UKR	Males	0.9462	0.9586	0.8391

D.5 Estimates across reference registers

Figure S19 reports the estimated unauthorised population (in thousands) and Figure S20 the same estimates as a share of the reference population ($\hat{\xi}/N$), in both cases by reference register (PESEL, Tax, ZUS), origin (Ukrainian vs non-Ukrainian), and estimation method (Poisson PMLE, NB-MLE). In recent years the Ukrainian share is much smaller than the non-Ukrainian share, because the 2022 Temporary Protection Directive expanded the Ukrainian reference population N without a matching increase in the unauthorised count m.

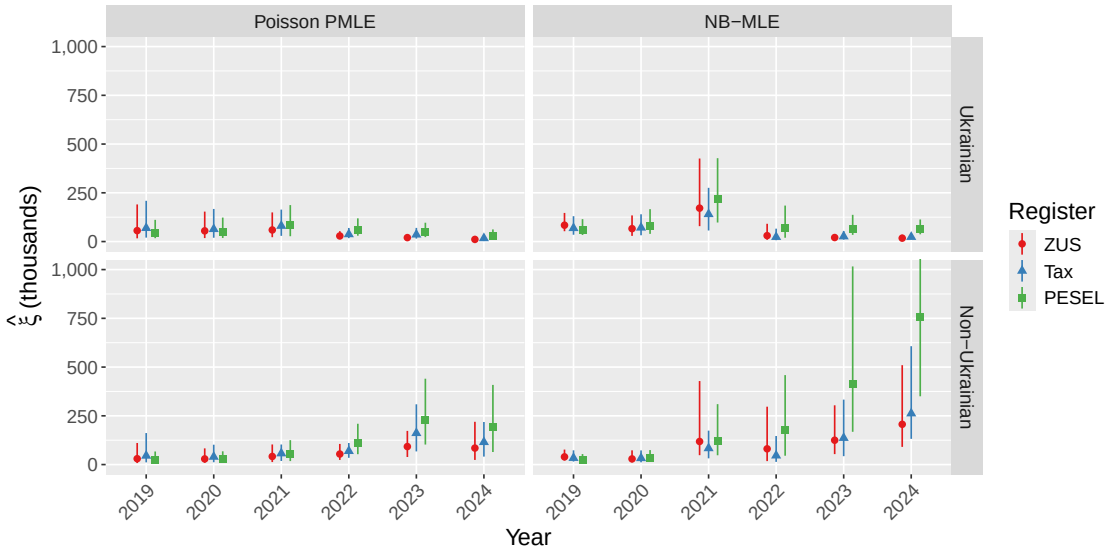


Figure S19: Estimated unauthorised population (in thousands) by reference register (PESEL, Tax, ZUS), origin (Ukrainian vs non-Ukrainian), and estimation method (Poisson PMLE, NB-MLE). Points are cluster fractional-weighted bootstrap medians with 95% percentile confidence intervals ($R = 199$, clustered by country); values are truncated at one million.

Under Poisson, the shares are similar across registers. The Ukrainian share falls from about 8–13% in 2019 to about 1% in 2024, following the post-2022 growth of the denominator. The non-Ukrainian share is about 14–27% through 2022 and 17–30% afterwards, with confidence intervals that overlap across the three registers.

Under NB, several non-Ukrainian point estimates exceed 50% of the reference population (for example PESEL at 79% and ZUS at 61% in 2024), and some upper confidence bounds exceed 100% and are truncated in the figure. These values follow from the γ–θ confounding discussed in Section F.1, which inflates $\hat{\alpha}$ and hence $\hat{\xi}$ under the NB variance function. We use Poisson as the primary estimator and NB as a sensitivity bound.

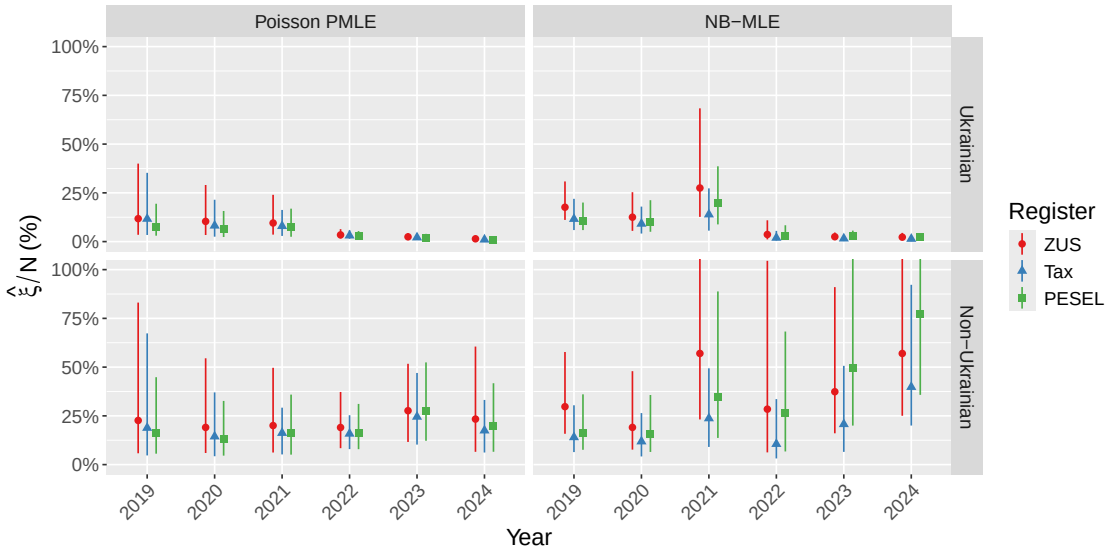


Figure S20: Estimated unauthorised population as a share of the reference population ($\hat{\xi}/N$), by register, origin, and estimation method. Points are cluster fractional-weighted bootstrap medians with 95% percentile confidence intervals ($R = 199$, clustered by country); shares are truncated at 100%.

D.6 Estimates across different auxiliary counts (n)

We refit S8 against the ZUS reference, replacing n with each of three counts: all adult police records (police total), the subset whose subject held a PESEL number (police PESEL ID), and prison records. The three datasets share the same m, N and cells, so any change in $\hat{\xi}$ comes from the auxiliary source alone.

Table S15 reports how informative each n is. All three correlate positively with m, but the Spearman rank correlation falls from 0.75 (police total) to 0.63 (prison). The detection slope β is identified only from cells with $m_{it}n_{it} > 0$, and the zero share rises from $30\%/69\%$ (males/females) for police total to $58\%/91\%$ for prison. With 91% of female prison cells empty, the small-counts destabilise NB-MLE (Section F.1).

Poisson PMLE is insensitive to this choice. Its yearly trajectory is stable across the three counts, broadly overlapping 80% and 95% percentile intervals (Figure S21), and the same agreement holds by Ukrainian origin (Figure S22). NB-MLE under all three counts it reproduces the implausible 2021 and 2024 spikes, so we report it only as a sensitivity bound. The one visible exception is the PESEL-ID estimate for 2021, which drops across all groups.

Table S15: Auxiliary detection counts used in the S8 sensitivity check: Pearson and Spearman correlation with the apprehension count m across community–year–sex cells, and the share of cells with a zero auxiliary count by sex.

Auxiliary count (n)	Pearson $r(m,n)$	Spearman $\rho(m,n)$	Zeros, Females (%)	Zeros, Males (%)
Police (total)	0.71	0.75	69	30
Police (PESEL ID)	0.53	0.73	79	37
Prison	0.66	0.63	91	58

Table S16: Model fit and estimated population size under S8 with three auxiliary detection counts. AIC, BIC and QAIC for Poisson PMLE, AIC and BIC for NB-MLE; $\hat{c}$ is the Pearson dispersion of each Poisson PMLE fit and QAIC $= -2\ell/\hat{c} + 2(K+1)$ uses the dispersion of the primary fit, police total ($\hat{c} = 9.0$), for all three sources. $\hat{\xi}$ is the sum of the yearly point estimates (in thousands). Lower NB-MLE AIC/BIC reflects over-dispersion accommodation and is not grounds for preferring it (Section 5.2).

	Poisson PMLE				NB-MLE		$\hat{\xi}$ (000s)	
Auxiliary count (n)	AIC	BIC	QAIC	$\hat{c}$	AIC	BIC	Po	NB
Police (total)	11,176	11,280	1,273	9.0	5,186	5,295	560	948
Police (PESEL ID)	12,954	13,058	1,470	13.7	5,374	5,483	473	761
Prison	13,241	13,345	1,502	10.8	5,288	5,398	552	878

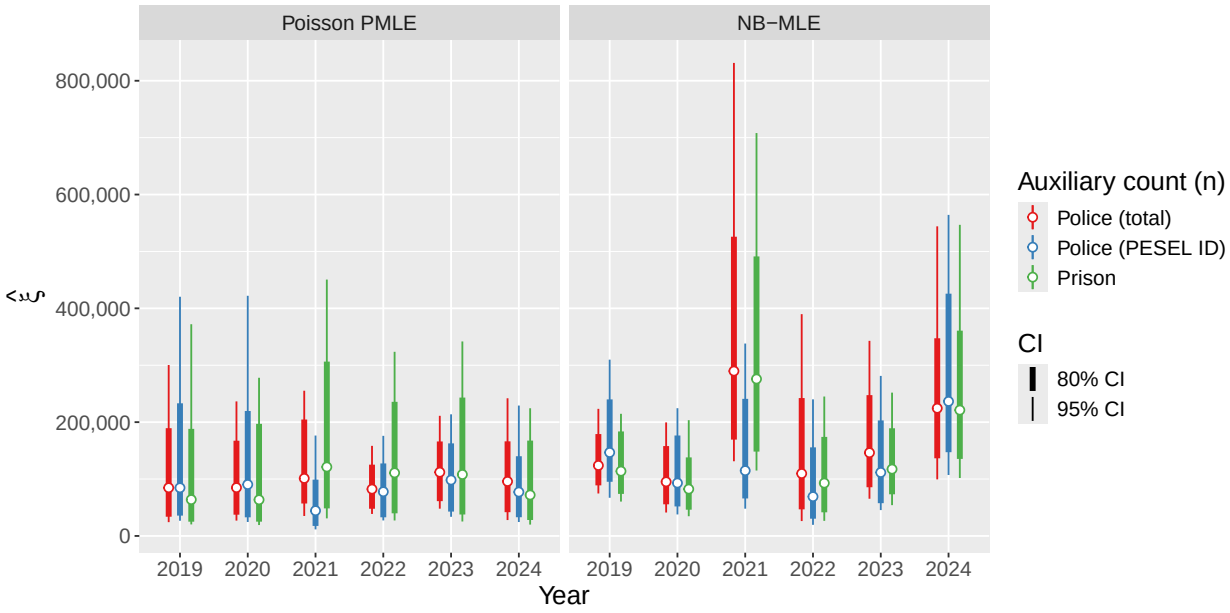


Figure S21: Estimated population $\hat{\xi}$ by year under S8, refitted with three auxiliary detection counts n, by estimation method. The counts are: police total (all adult police records in criminal or registration proceedings), police PESEL ID (the subset whose subject held a PESEL number), and prison (prison records). Points are cluster fractional-weighted bootstrap medians; thick bars give 80% and thin bars 95% percentile confidence intervals (199 replicates, clustered by country).

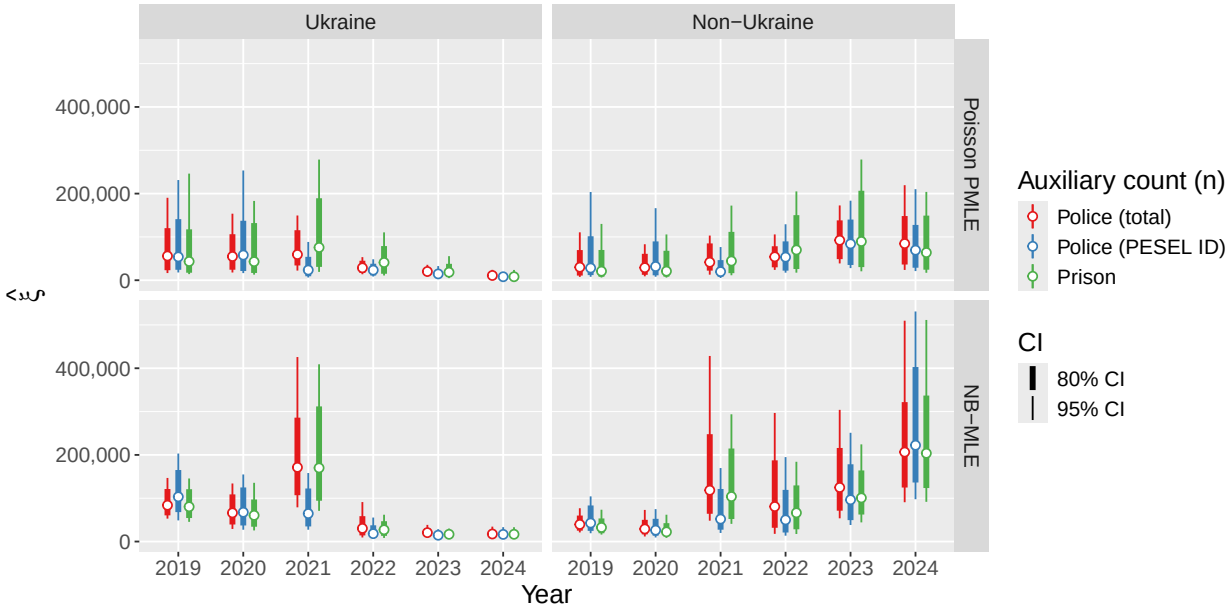


Figure S22: As Figure S21, by Ukrainian versus non-Ukrainian origin.

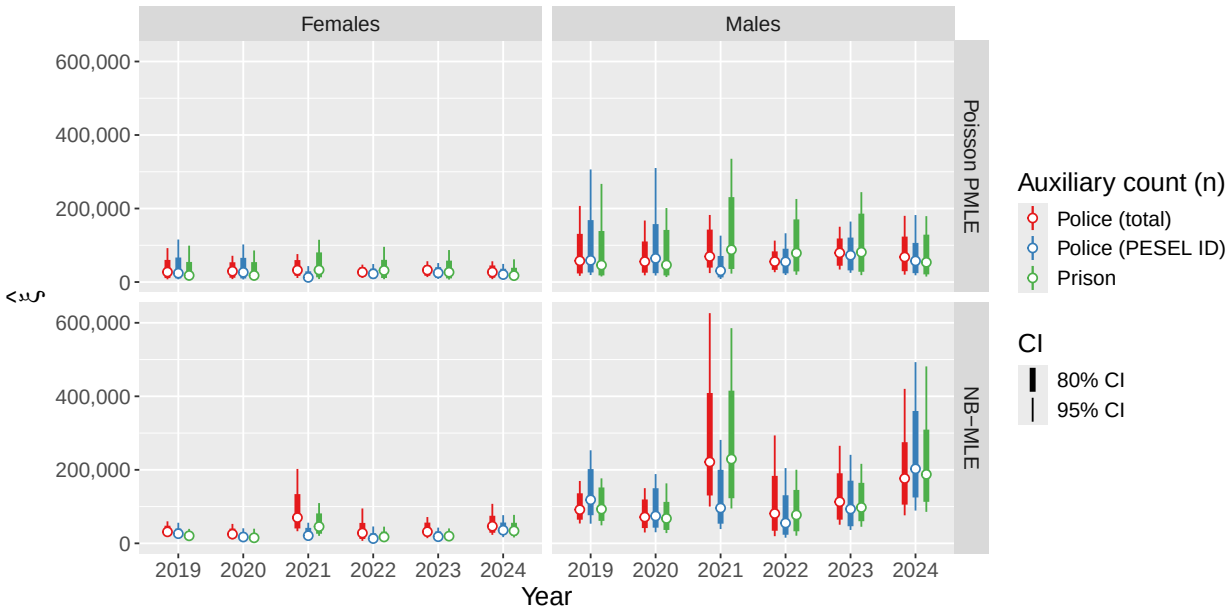


Figure S23: As Figure S21, by sex. The prison count is zero in 91% of female cells, which inflates the female intervals.

E Model diagnostics for S8 specification

E.1 Country-level leave-one-out sensitivity for the S8 specification

Figure S24 shows leave-one-out sensitivity by country of origin under S8: each bar is the change in $\hat{\xi}$ when that country is dropped. Ukraine is excluded, because dropping it removes the entire UKR stratum and leaves the year-by-UKR interaction unidentifiable. Under Poisson, leverage is concentrated in Belarus ($+64\%$) and Georgia (-49%), with no other country above about 6%. Under NB the effects are smaller and more spread out, the largest being the unknown/stateless group ($+13\%$), Vietnam (-9%), and Georgia (-8%). Cluster bootstrap SE takes this into account.

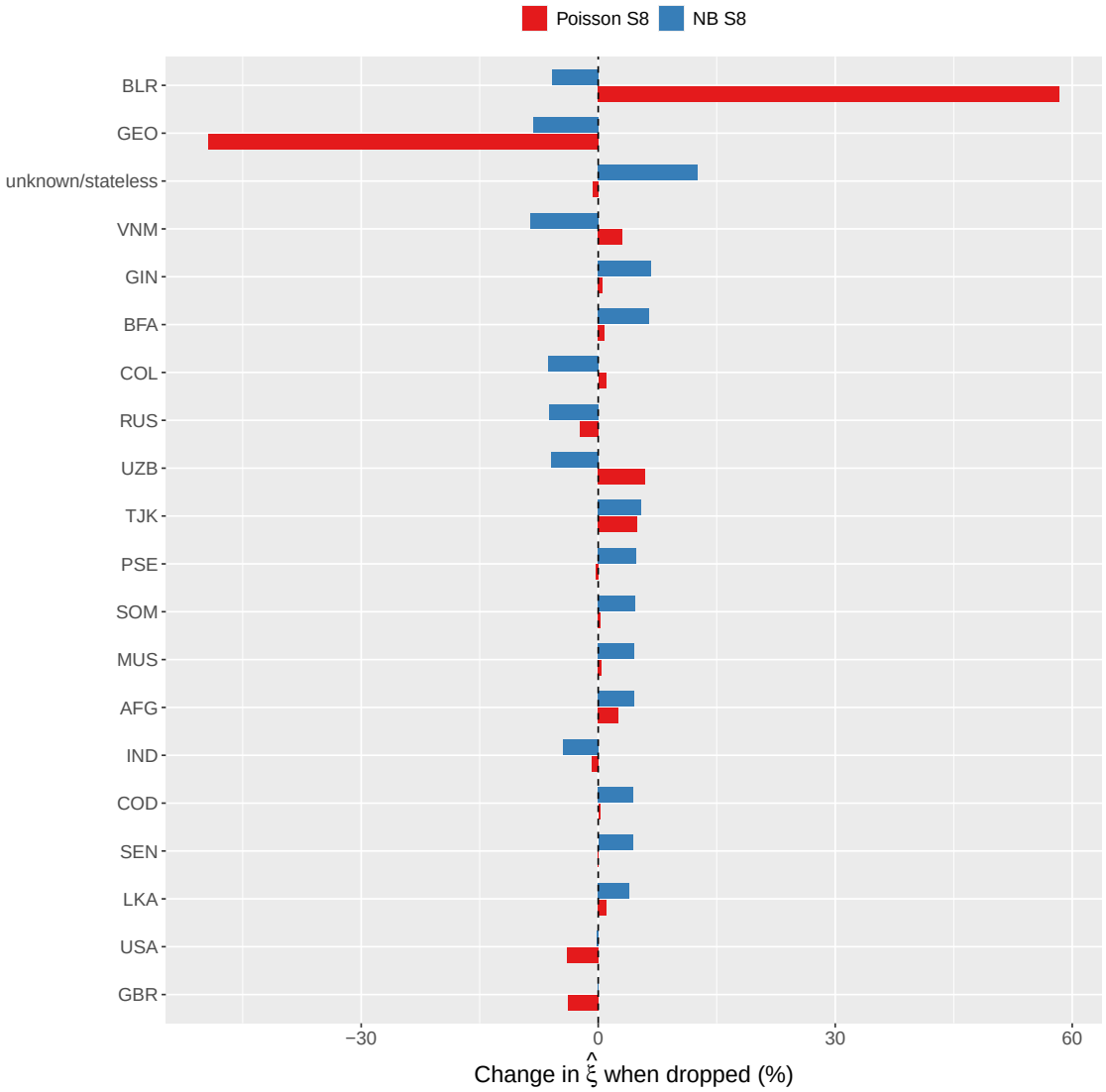


Figure S24: Leave-one-out sensitivity by country of origin under S8: change in $\hat{\xi}$ when each country is dropped, top 20 shown (Ukraine excluded). Poisson is dominated by a few origins; NB influence is smaller and more evenly spread, reflecting its quadratic variance function.

E.2 Observation-level influence analysis

We additionally assess influence at the observation level, dropping each country-by-year-by-sex cell individually.

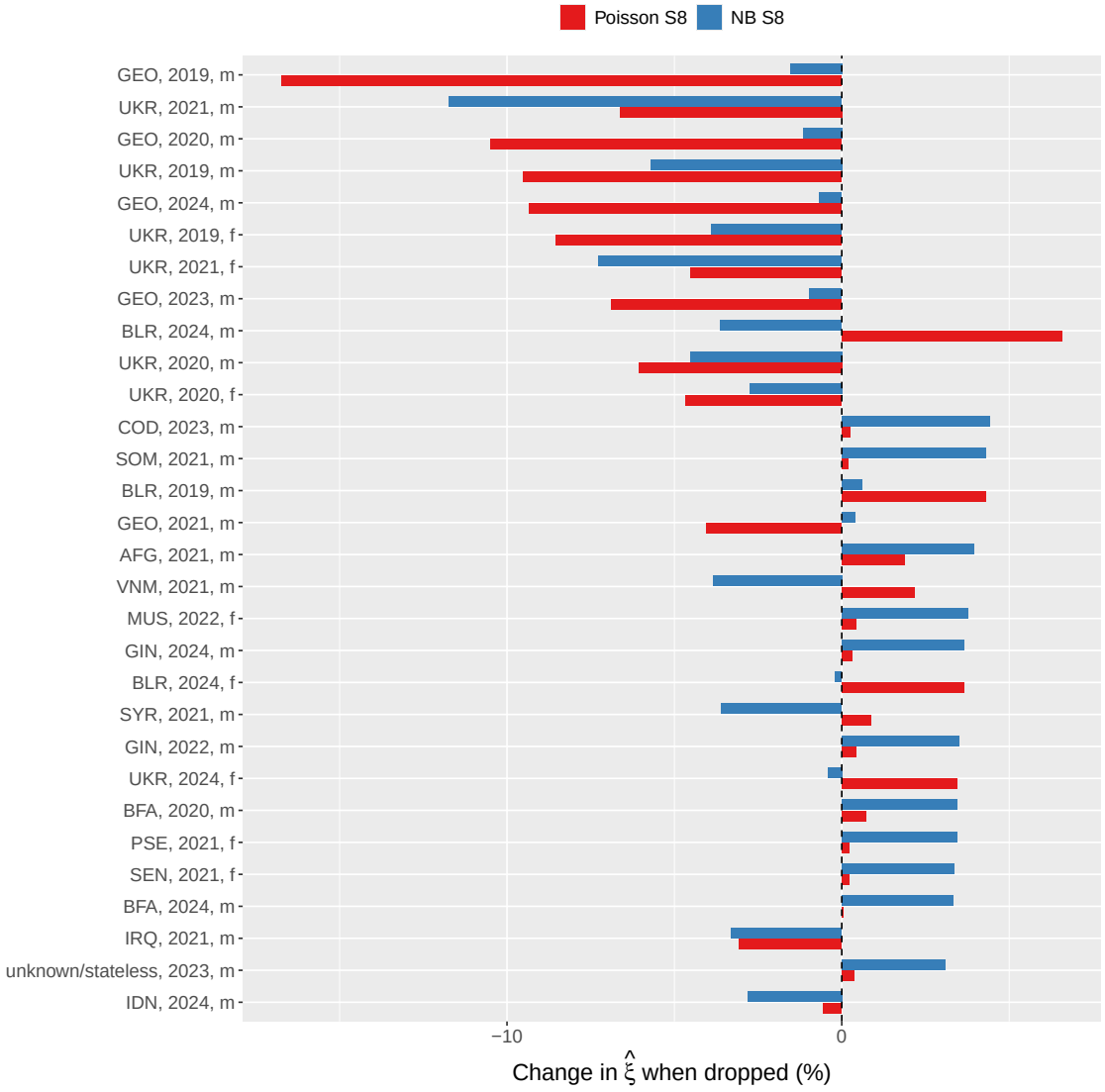


Figure S25: Observation-level LOO sensitivity (S8), top 30 cells. Under Poisson, influence tracks μ (large cells dominate); under NB, the variance function $V(\mu) = \mu + \mu^2/\theta$ decrease large cells and lifts small-μ outliers.

Figure S25 ranks individual cells by their influence on $\hat{\xi}$. Under Poisson the leaders are Georgian and Ukrainian male cells with high apprehension counts (e.g. Georgia 2019, -17.5%). Under NB the largest single cell is Ukraine 2021 (-12.2%), but the variance function down-weights the large Georgian cells (Georgia 2019 falls to -1.4%) and lifts small-μ origins such as Congo, Somalia, and Mauritius that carry almost no Poisson influence. Figure S26 plots the two rankings against each other to make this asymmetry visible.

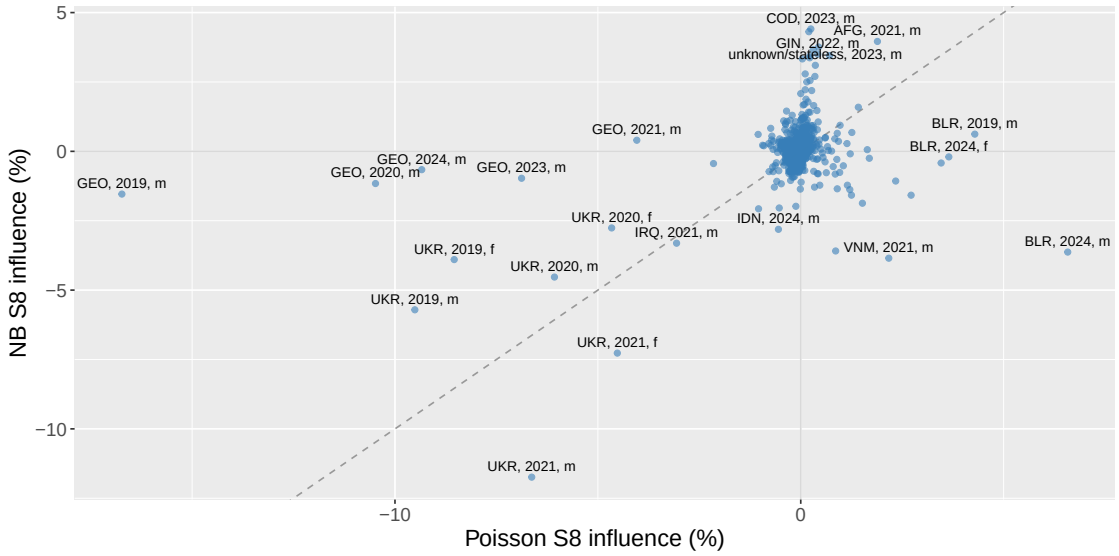


Figure S26: Observation-level LOO: Poisson vs NB influence. Points off the diagonal are cells that affect the two models asymmetrically; small-μ cells with high apprehension rates sit in the upper-left (influential for NB, not Poisson), reflecting the γ–θ confounding from the simulation study.

Under Poisson, the likelihood weight of cell i scales with μ_i, so large cells (high N and m) dominate. Under NB, the variance function $V(\mu) = \mu + \mu^2/\theta$ decrease the weight of high-μ cells and gives relatively more leverage to small-μ cells (low N but $m > 0$). This reflects the γ–θ confounding identified in the simulation study (Section F): small-μ cells are where γ is hardest to separate from overdispersion.

E.3 Anscombe residuals vs fitted values under S8

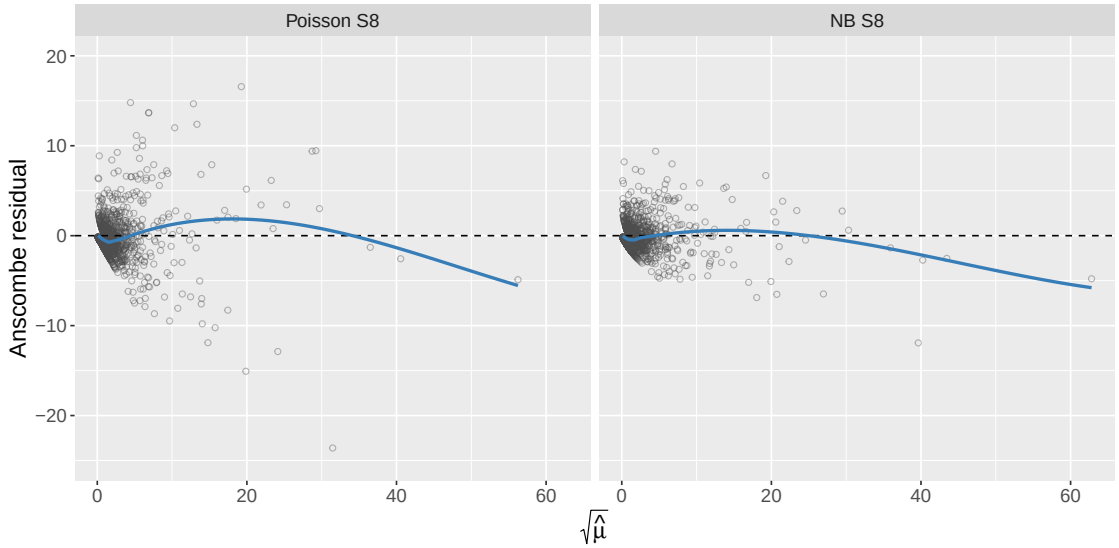


Figure S27: Anscombe residuals vs fitted values under S8

E.4 Gamma profiles for S8

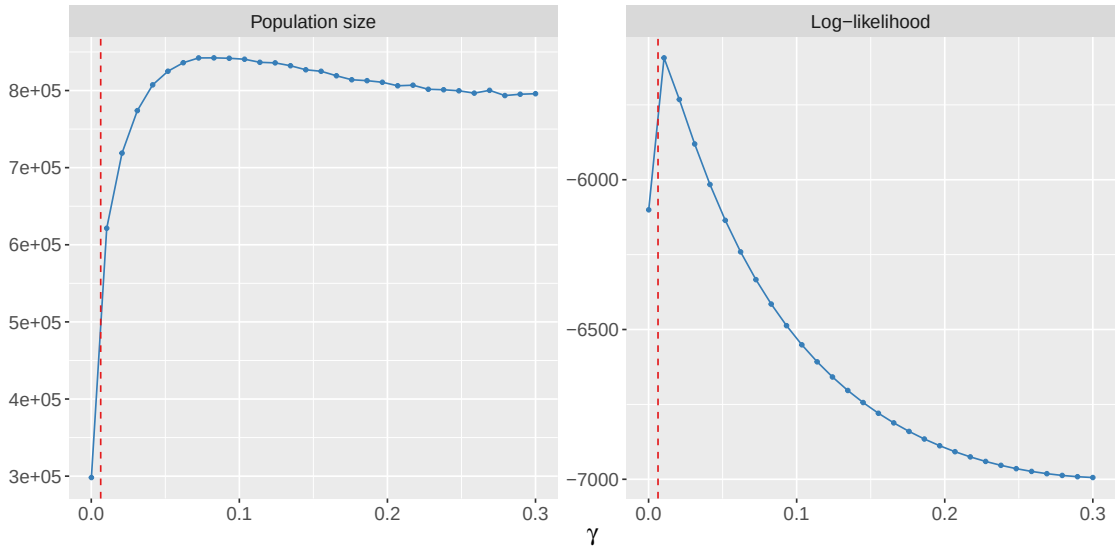


Figure S28: Poisson S8 gamma profile

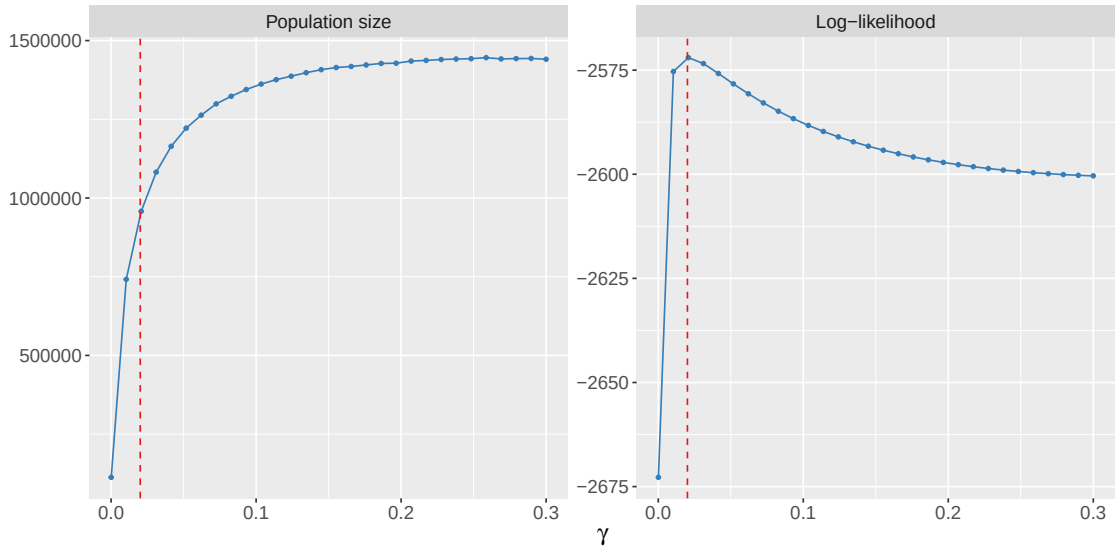


Figure S29: NB-MLE S8 gamma profile

E.5 Sensitivity to covariate-varying gamma

The baseline specification uses a single estimated γ for all observations. We examine whether allowing γ to vary by year, sex, and/or Ukrainian origin improves fit, fixing $\alpha = \text{year} \times \text{UKR} + \text{sex}$ and $\beta = \text{year}$ (S8) and varying only the gamma specification.

Table S17: AIC, BIC and QAIC by gamma specification and estimation method. All models use the S8 specification for alpha and beta with the ZUS reference population.

	Poisson PMLE					NB-MLE		
γ specification	AIC	BIC	QAIC	$\hat{c}$	k_γ	AIC	BIC	k_γ
constant	11,176.0	11,280.2	1,273.3	9.04	1	5,186.0	5,295.4	1
$\gamma \sim$ year	10,976.4	11,106.6	1,260.1	9.71	6	5,179.3	5,314.8	6
$\gamma \sim$ sex	11,169.3	11,278.7	1,274.3	8.95	2	5,186.5	5,301.1	2
$\gamma \sim$ UKR	11,151.2	11,260.6	1,272.3	8.98	2	5,187.9	5,302.5	2
$\gamma \sim$ year + UKR	10,935.5	11,071.0	1,257.4	9.72	7	5,181.3	5,322.0	7
$\gamma \sim$ year + sex	10,816.4	10,951.9	1,244.2	9.00	7	5,178.9	5,319.6	7
$\gamma \sim$ year + sex + UKR	10,763.6	10,904.3	1,240.1	8.80	8	5,180.9	5,326.8	8

Note: S8 specification: $\alpha \sim$ year $\times$ UKR + sex, $\beta \sim$ year. ZUS reference population. k_γ = number of gamma parameters; $\hat{c}$ = Pearson dispersion of the Poisson PMLE fit; QAIC $= -2\ell/\hat{c}_{S8} + 2(K+1)$ with $\hat{c}_{S8}$ from the adopted specification, constant γ ($\hat{c}_{S8} = 9.0$). Lower AIC/BIC/QAIC is better.

Letting γ vary by covariates improves the Poisson fit substantially (ΔAIC ≈ 440 for γ by year, sex and Ukrainian origin) but barely changes the Negative Binomial (ΔAIC ≈ 3). This is the γ–θ trade-off: under NB, the dispersion parameter θ already absorbs the detection heterogeneity that covariate-varying γ captures under Poisson. The NB θ stays close to 1.43 across all gamma specifications, so γ and θ act as substitutes.

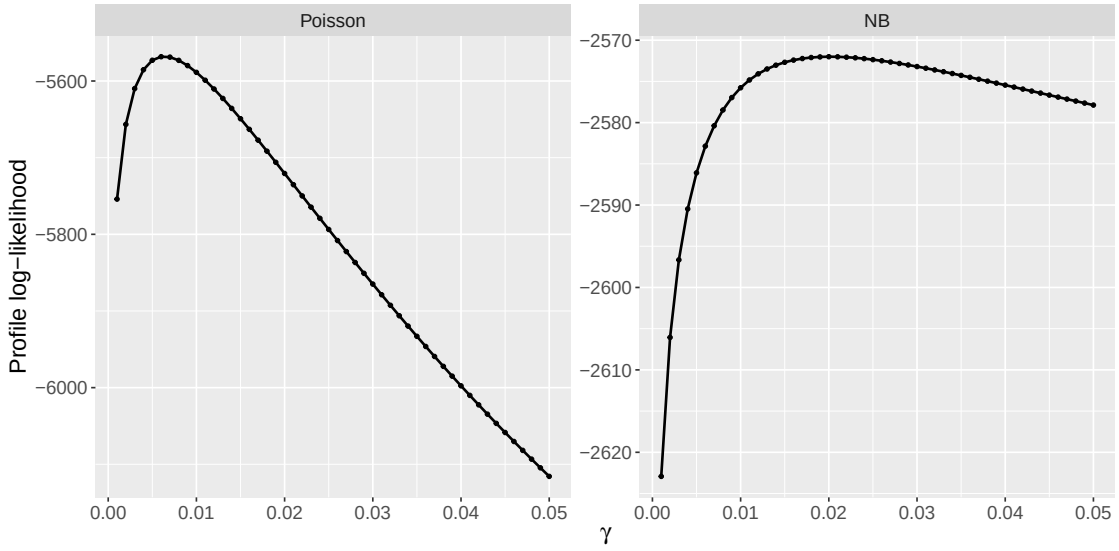


Figure S30: Profile log-likelihood of the constant γ: Poisson (left, sharp peak) vs NB (right, flat). The flat NB profile reflects weak identification from γ–θ confounding.

We keep constant γ in the main results because it is well-identified under Poisson, leaves the NB fit (and hence the Poisson–NB comparison) unchanged, and the resulting changes in population size

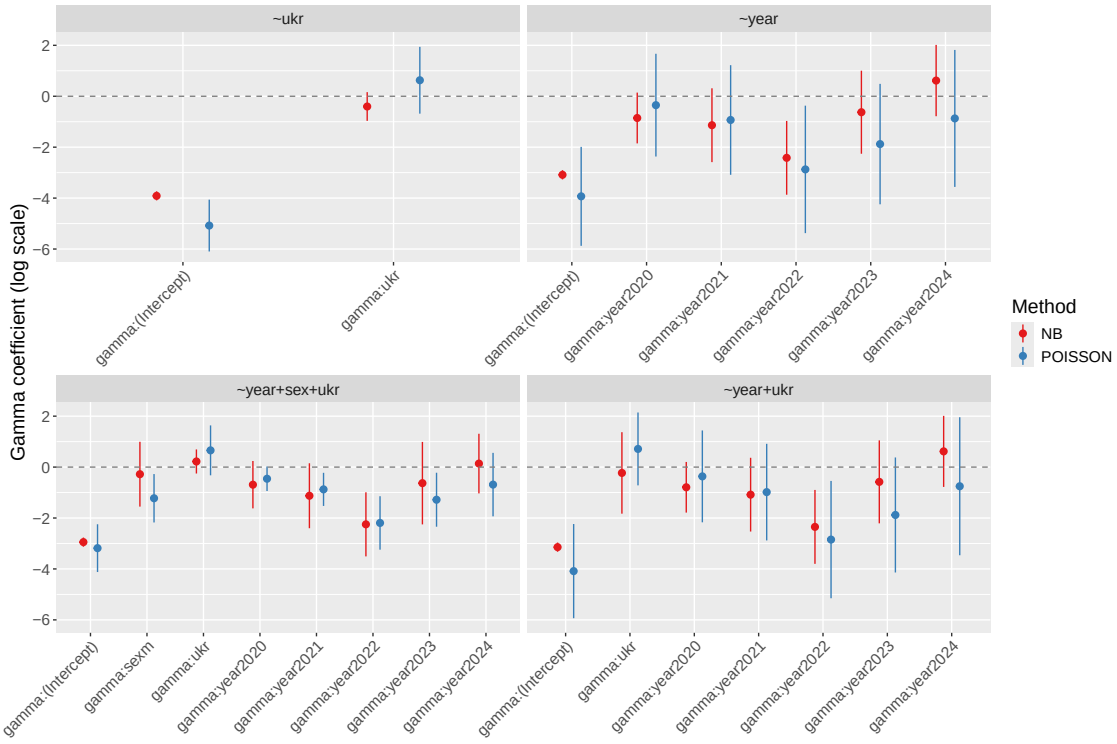


Figure S31: Gamma coefficients (log scale) with 95% Wald CIs. Under NB the intervals are wide and include zero, reflecting weak identification.

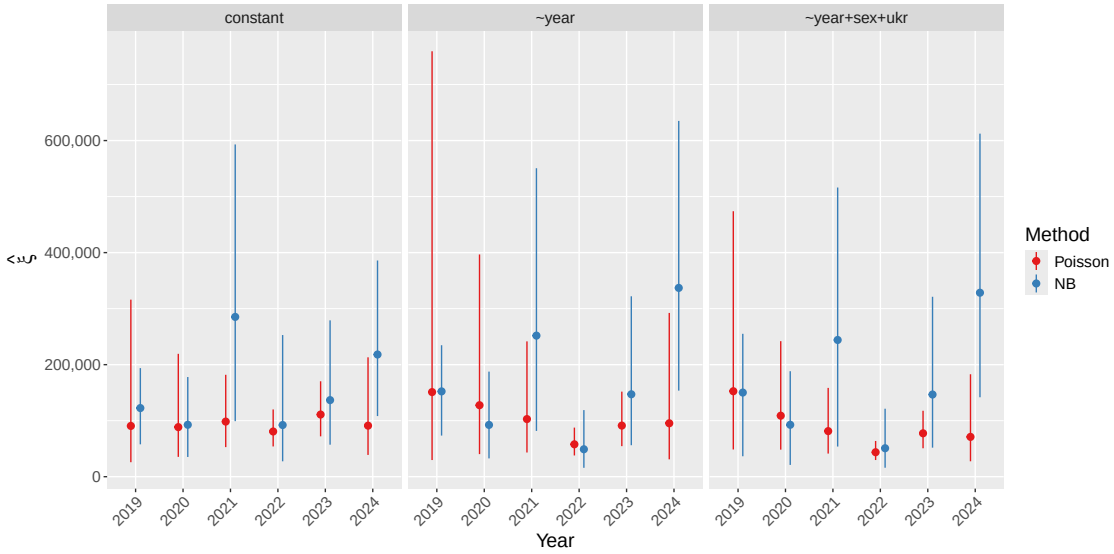


Figure S32: Estimated unauthorised population by year: constant vs covariate-varying γ (delta-method 95% CI). The covariate-varying estimates fall within the bootstrap intervals of the baseline model.

fall within the bootstrap intervals.

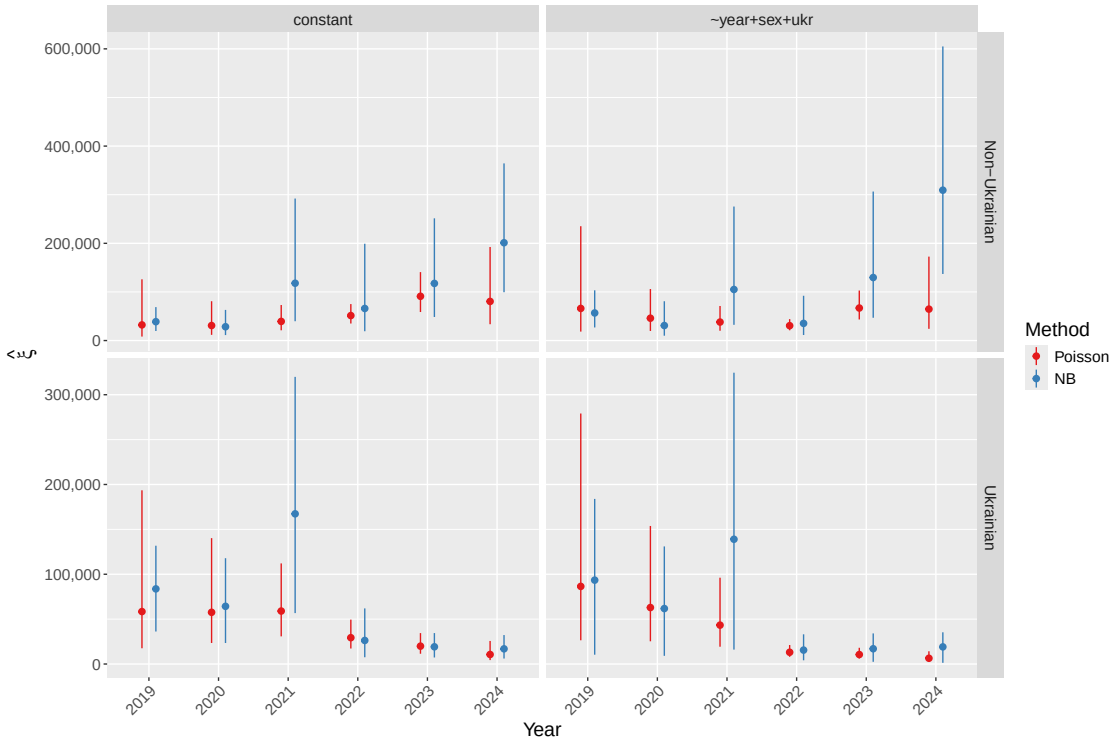


Figure S33: Estimated unauthorised population by Ukrainian status: constant vs covariate-varying γ (delta-method 95% CI).

E.6 Profile log-likelihood of α and β

To assess identification of the structural parameters α and β under both Poisson and NB, we compute concentrated profile log-likelihoods with `uncounted::profile_alpha()` and `uncounted::profile_beta()`, varying each intercept over a grid while holding the other coefficients at their MLE.

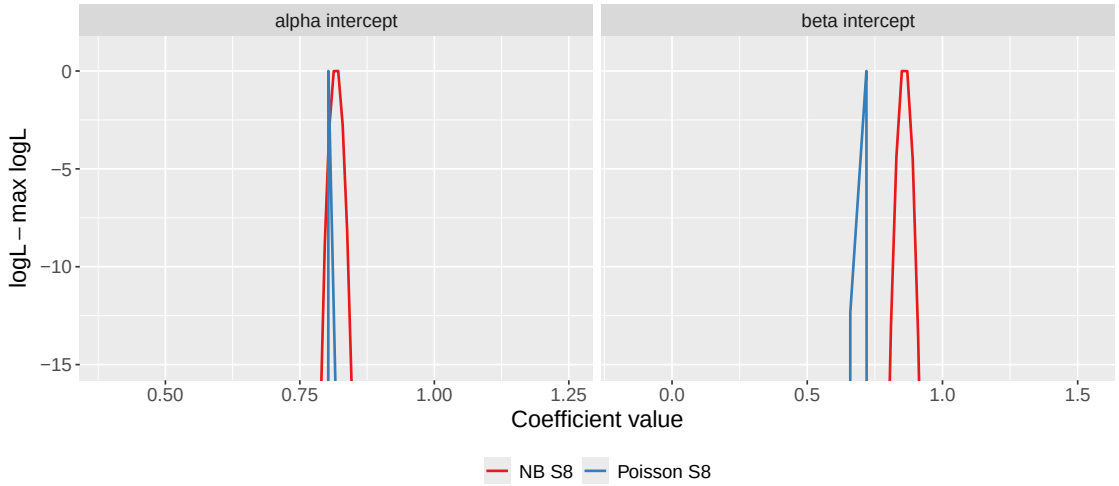


Figure S34: Concentrated profile log-likelihood of the α intercept (left) and β intercept (right) under Poisson S8 and NB S8. Both parameters are sharply identified under both models, unlike γ, whose NB profile is flat (see main text).

Both α and β intercepts are sharply identified under Poisson and NB, with profile curvatures of similar magnitude. This contrasts with γ, whose NB profile is essentially flat. The γ–θ confounding is therefore specific to the baseline-detection parameter and does not extend to the structural community-anchor and exposure-elasticity parameters.

F Simulation studies

F.1 Details about the simulation

We assess the properties of estimation methods by simulations motivated from the 2024 Polish data. The main objective is to choose the suitable estimator for the application later.

For each replication, we generate (N_i, n_i, m_i) independently across the communities, where the number of communities is fixed at 100. First, let $N_i \sim \text{LogNormal}(5.5, 2.75)$, truncated to $N_i \geq 3$. Next, generate n_i from either the Poisson or NB distribution, where $\log \mathbb{E}(n_i \mid N_i) = c_0 + c_1 \log N_i$, and the parameters of either distribution are derived from the police data n_i and the ZUS register N_i — see Section 4.2.3 for details of these data sources. Finally, the mean of m_i is given by (4) and (5) with $(\alpha, \beta, \gamma) = (0.7, 0.5, 0.005)$, and m_i follows either the Poisson or NB distribution, derived from the Polish data — see Supplementary Materials Section F for the details. Notice that the dispersion of m_i under repeated simulation of (N_i, n_i, m_i) in this way is actually larger than that observed in the 2024 Polish data, because (N_i, n_i) are not constants in the simulation.

Crossing the two data-generating processes of n_i and m_i yields four scenarios: Po(n)–Po(m), Po(n)–NB(m), NB(n)–Po(m), NB(n)–NB(m). In addition, we consider two estimation setups.

- In Setup-I, we apply regrouping to create an ad hoc 'rest' community from all the data points with $m_i n_i = 0$, such that the detection intensity is modeled without the offset, i.e. $\gamma \equiv 0$ in (5). Notice that the mean structure of Poisson PMLE or NB-MLE is misspecified in this setup, since all the data are generated with $\gamma = 0.005$.
- In Setup-II, we allow for non-trivial offset $\gamma > 0$ in (5), such that the communities with $m_i n_i = 0$ are used directly in estimation. Notice that the mean structure of Poisson PMLE or NB-MLE is correct in this setup, but neither the Poisson nor the NB likelihood is correct in any scenario since (N_i, n_i) are not constants in the simulation.

Table S18: Simulation results, 2000 replications. Bias relative to $\xi = \sum_{i=1}^{100} \xi_i$. RMSE, root mean squared error. Coverage of ξ. Confidence interval (CI) of ξ by (10), CI width relative to ξ.

	Setup-I: $\gamma = 0$, regrouped 'rest' community to avoid $m_i n_i = 0$				
Data	Estimator	Bias (%)	RMSE	Coverage (%)	CI width (%)
Po(n)-Po(m)	Po-PMLE	-0.4	7,599	95.7	54.2
Po(n)-Po(m)	NB-MLE	-2.1	7,752	93.5	47.4
NB(n)-NB(m)	Po-PMLE	11.6	103,375	79.6	253.3
NB(n)-NB(m)	NB-MLE	-31.1	19,604	70.6	112.5
	Setup-II: $\gamma > 0$, communities with $m_i n_i = 0$ allowed				
Data	Estimator	Bias (%)	RMSE	Coverage (%)	CI width (%)
Po(n)-Po(m)	Po-PMLE	-0.4	3,899	95.1	40.3
Po(n)-Po(m)	NB-MLE	-3.9	9,076	77.4	33.7
NB(n)-NB(m)	Po-PMLE	70.6	144,046	88.2	422.5
NB(n)-NB(m)	NB-MLE	10.9	24,405	91.8	178.6

Some simulation results based on 2000 replications are highlighted in Table S18; see Supplementary Materials Section F.2–F.4 for the full results. Both the Poisson PMLE ("Po-PMLE" in Table S18) and the NB-MLE are calculated for each data generation scenario and estimation setup.

Consider the Poisson PMLE. First, given Po(n)–Po(m), it is essentially unbiased in Setup-II and has the correct coverage at the 95% level; although the mean structure is misspecified in Setup-I, it causes little bias given an offset of the magnitude $\gamma = 0.005$, such that the larger RMSE in Setup-I is rather due to the loss of efficiency from regrouping to create the 'rest' community.

Next, given NB(n)–NB(m), Poisson PMLE does suffer from bias as well as a very large variance (either in terms of the CI width or RMSE). This serves as a warning that, despite the asymptotic consistency, the performance of Poisson PMLE can deteriorate greatly given small sample size, such as 100 here, and excessive over-dispersion, such as by LogN(N)-NB(n)–NB(m).

Consider now the NB-MLE. First, given Po(n)–Po(m), it is biased despite the correct mean structure in Setup-II: in fact, the low coverage of 77.4% suggests that the bias is relatively large compared to the standard error. It suffers a loss of efficiency in Setup-I due to regrouping, just like Poisson PMLE. Meanwhile, its bias is somewhat reduced from -3.9% in Setup-II to -2.1% Setup-I, which suggests that the sensitivity of estimating γ (in Setup-II) can possibly be more damaging to NB-MLE than mean-mis-specification (in Setup-I) which avoids the need to estimate γ.

Next, given NB(n)–NB(m), NB-MLE also suffers from bias but with a much smaller variance than Poisson PMLE; its bias is smaller in Setup-II (10.9% vs -31.1%) yet still large enough to lower the coverage, reflecting the sensitivity of estimating γ. This sensitivity is most severe when the data carry no over-dispersion: under Po(n)–Po(m), NB-MLE overestimates $\gamma = 0.005$ by a factor of 24 ($\hat{\gamma}$ averages 0.12, hitting the upper boundary in 16% of replications).

In summary, these results point to the following choices for the application to the Polish data: (i) allow offset $\gamma > 0$ to avoid ad hoc regrouping of data over time, and the associated loss of efficiency; (ii) adopt Poisson PMLE as the primary estimator, but be aware of any excessive over-dispersion; (iii) NB-MLE is unstable due to offset-Gamma confounding and small-counts sensitivity, although it can still provide sensitivity checks with or without non-trivial offset γ, i.e. whether the NB-MLE is included in the confidence intervals based on the Poisson PMLE.

We conduct two simulation studies to assess the finite-sample properties of the $\hat{\xi}$ estimator under known data-generating processes (DGPs). The summary in Section F.1 gives the main structure; here we report the full parameter values and the calibration procedure, along with both simulation studies in detail.

F.2 Simulation 1: effect of aggregation

In the first simulation ($R = 2{,}000$ replications, $C = 100$ countries per replicate) we aggregate the countries that violate the positivity condition ($n > 0$, $m > 0$, $n/N < 1$) into a single "rest" group, which tests the aggregation procedure described in the main text.

Table S19: Simulation 1: bias, RMSE, coverage and CI width across four DGP scenarios under aggregation (all $m > 0$, $n > 0$). Analytical (delta-method) CIs. $R = 2{,}000$ replications.

Model	Bias (%)	Bias BC (%)	RMSE	Coverage (%)	CI width (%)	R_{ok}
NB(n)-NB(m)						
NB	-31.1	-39.0	19,604	70.6	112.5	1950
NB (c)	-31.1	-38.2	19,605	70.9	113.4	1950
NLS	-47.9	-53.3	21,494	81.8	766.4	1950
OLS	-34.0	-43.9	20,499	74.4	127.2	1950
Poisson	11.6	11.2	103,375	79.6	253.3	1950
Poisson (c)	11.4	4998.1	101,745	79.5	253.5	1950
NB(n)-Po(m)						
NB	-39.9	-40.7	21,184	40.9	57.8	1292
NB (c)	-39.9	-40.7	21,184	41.2	57.8	1292
NLS	-34.5	-34.7	8,897	56.6	66.7	1292
OLS	-21.8	-24.9	12,835	83.0	82.4	1292
Poisson	-31.6	-31.9	16,643	46.4	52.1	1292
Poisson (c)	-31.6	-31.8	16,643	46.5	52.1	1292
Po(n)-NB(m)						
NB	-15.8	-26.6	20,216	82.7	144.1	1974
NB (c)	-15.8	-25.4	20,219	82.7	145.8	1974
NLS	-32.9	-47.9	26,258	83.9	4587.2	1971
OLS	-17.6	-30.6	21,171	87.1	160.0	1974
Poisson	155.0	153.5	1,368,824	83.4	490.3	1974
Poisson (c)	103.3	24749.5	218,537	82.8	481.5	1974
Po(n)-Po(m)						
NB	-2.1	-2.8	7,752	93.5	47.4	1465
NB (c)	-2.1	-2.7	7,752	93.4	47.5	1465
NLS	9.3	9.0	3,612	92.2	86.2	1465
OLS	7.7	3.8	11,130	96.1	116.0	1465
Poisson	-0.4	-1.0	7,599	95.7	54.2	1465
Poisson (c)	-0.4	-0.9	7,599	95.7	54.3	1465

Note: bias and CI width as a percentage of true ξ; RMSE on the original scale; coverage is the share of replications whose 95% analytical CI contains ξ; R_{ok} is the number of finite replications out of 2,000. NLS bias, bias BC and RMSE use median-based summaries (root-median-squared error) because a few replicates are numerically extreme under NB-generated data; the other estimators use mean-based summaries.

Under the correctly specified scenario Po(n)-Po(m), the Poisson estimator is essentially unbiased (-0.4%) with near-nominal coverage (95.7%). Under the misspecified NB(n)-NB(m) scenario it shows moderate positive bias (about 12%) but keeps reasonable coverage (about 80%). The NB estimator is biased downward under aggregation (about -31%), because the "rest" group absorbs small-country variation, and OLS is substantially biased downward (about -34%) through the $\log(m + 1)$ transformation. NLS is numerically unstable under NB-generated data, so its bias and RMSE in Table S19 use median-based summaries (see the note); even so, it shows large negative bias under NB(n)-NB(m) and Po(n)-NB(m). The mixed scenario NB(n)-Po(m) is the most challenging for all estimators: Poisson shows -31.6% bias and only 46% coverage, reflecting the effect of overdispersed auxiliary counts on the detection rate.

F.3 Simulation 2: zeros in the counts

The second simulation allows zeros in both m and n (no aggregation), testing the ability of the γ offset to handle sparse data. Bootstrap confidence intervals ($R_{\text{boot}} = 499$, no clustering) are also computed.

Table S20: Simulation 2: bias, RMSE, coverage and CI width with zeros allowed (γ estimated). Analytical CIs. $R = 2{,}000$ replications.

Model	Bias (%)	Bias BC (%)	Bias med (%)	RMSE	Coverage (%)	CI width (%)	R_{ok}
NB(n)-NB(m)							
NB	10.9	-0.6	10.0	24,405	91.8	178.6	2000
NB (c)	10.9	1.2	10.0	24,418	91.4	181.2	2000
NLS	-7.6	-21.9	-5.0	27,405	89.2	4599.0	1979
OLS	-80.5	-81.6	-80.8	35,348	0.0	27.3	2000
Poisson	70.6	69.8	56.5	144,046	88.2	422.5	2000
Poisson (c)	66.4	74.8	55.4	117,479	88.0	420.4	2000
NB(n)-Po(m)							
NB	0.8	0.3	0.9	4,412	91.8	34.7	2000
NB (c)	0.9	0.4	0.8	4,418	91.8	34.7	2000
NLS	0.4	0.2	0.6	2,896	94.4	79.0	1984
OLS	-75.2	-75.7	-75.3	32,577	0.0	23.2	2000
Poisson	0.8	0.3	0.8	4,392	95.7	42.1	2000
Poisson (c)	0.8	0.4	0.8	4,392	95.7	42.1	2000
Po(n)-NB(m)							
NB	6.9	-4.7	5.4	20,226	92.8	178.8	2000
NB (c)	6.9	-2.9	5.4	20,222	92.8	181.5	2000
NLS	-41.2	-62.0	-40.4	25,595	84.6	3000.4	1980
OLS	-77.9	-79.0	-78.3	33,975	0.2	28.2	2000
Poisson	77.8	76.3	53.5	534,916	83.7	392.2	2000
Poisson (c)	63.5	64.4	50.2	165,647	83.8	391.3	2000
Po(n)-Po(m)							
NB	-3.9	-4.4	-0.6	9,076	77.4	33.7	2000
NB (c)	-0.3	-0.7	-0.6	3,465	93.1	34.7	2000
NLS	-6.8	-7.3	-6.6	3,886	93.0	84.7	1997
OLS	-70.3	-70.9	-70.7	31,600	0.0	22.9	2000
Poisson	-0.4	-0.9	-0.7	3,899	95.1	40.3	2000
Poisson (c)	-0.3	-0.7	-0.6	3,413	95.3	40.4	2000

Note: bias and CI width as a percentage of true ξ; RMSE on the original scale. NLS bias, bias BC, bias med and RMSE use median-based summaries (root-median-squared error) under NB-generated data; other estimators use mean-based summaries.

With zeros allowed, OLS collapses (about 0% coverage) because $\log(m+1)$ attenuates the signal when most counts are small. NLS remains numerically unstable under NB-generated data; under median-based summaries it shows large negative bias under Po(n)-NB(m). Neither log-linearised estimator is suitable for sparse count data.

Under the correctly specified scenario Po(n)-Po(m), the Poisson estimator achieves -0.4% bias, RMSE of $3{,}899$, and 95.1% coverage. Under NB misspecification its bias rises to about 71–78% as overdispersion inflates the mean estimate, while coverage stays above 83% through wider CIs.

The NB estimator, which specifies $\text{Var}(m \mid X) = \mu + \mu^2/\theta$, achieves 7–11% bias with about 92% coverage under NB-generated data, against 71–78% for Poisson. The cost is γ–θ confounding: both θ and γ absorb variation beyond the Poisson mean, so γ is weakly identified. Table S22 shows that the unconstrained NB overestimates γ by a factor of 24 (mean $\hat{\gamma} = 0.12$ vs true 0.005) in 16% of Po(n)-Po(m) replications, hitting the optimisation boundary. RMSE rises from $3{,}413$ (Poisson constrained) to $9{,}076$ (NB unconstrained). The constrained NB (NB$_c$, $\alpha \in (0, 1)$) reduces boundary

Table S21: Simulation 2: analytical vs bootstrap CI coverage and width, by scenario and estimator.

CI	Model	Coverage (%)	CI width (%)
NB(n)-NB(m)			
Analytical	NB	91.8	178.6
Bootstrap	NB	93.3	190.0
Analytical	NB (c)	91.4	181.2
Bootstrap	NB (c)	93.0	190.5
Analytical	NLS	89.2	4599.0
Bootstrap	NLS	65.9	1023.8
Analytical	OLS	0.0	27.3
Bootstrap	OLS	0.0	22.3
Analytical	Poisson	88.2	422.5
Bootstrap	Poisson	75.8	201.5
Analytical	Poisson (c)	88.0	420.4
Bootstrap	Poisson (c)	75.4	200.5
NB(n)-Po(m)			
Analytical	NB	91.8	34.7
Bootstrap	NB	92.8	36.1
Analytical	NB (c)	91.8	34.7
Bootstrap	NB (c)	93.0	36.2
Analytical	NLS	94.4	79.0
Bootstrap	NLS	82.6	40.7
Analytical	OLS	0.0	23.2
Bootstrap	OLS	0.0	18.5
Analytical	Poisson	95.7	42.1
Bootstrap	Poisson	92.8	35.5
Analytical	Poisson (c)	95.7	42.1
Bootstrap	Poisson (c)	92.3	35.6
Po(n)-NB(m)			
Analytical	NB	92.8	178.8
Bootstrap	NB	93.6	185.8
Analytical	NB (c)	92.8	181.5
Bootstrap	NB (c)	93.6	185.6
Analytical	NLS	84.6	3000.4
Bootstrap	NLS	60.8	670.7
Analytical	OLS	0.2	28.2
Bootstrap	OLS	0.0	24.1
Analytical	Poisson	83.7	392.2
Bootstrap	Poisson	71.8	193.6
Analytical	Poisson (c)	83.8	391.3
Bootstrap	Poisson (c)	71.8	193.3
Po(n)-Po(m)			
Analytical	NB	77.4	33.7
Bootstrap	NB	92.8	36.4
Analytical	NB (c)	93.1	34.7
Bootstrap	NB (c)	92.6	35.7
Analytical	NLS	93.0	84.7
Bootstrap	NLS	87.2	52.0
Analytical	OLS	0.0	22.9
Bootstrap	OLS	0.0	19.7
Analytical	Poisson	95.1	40.3
Bootstrap	Poisson	92.8	36.1
Analytical	Poisson (c)	95.3	40.4
Bootstrap	Poisson (c)	93.1	35.5

Note: coverage is the share of replications whose 95% CI contains ξ; CI width is the median width as a percentage of ξ. Bootstrap CIs use $R_{\text{boot}} = 499$ per replicate without clustering. NLS CI widths can be very large under NB-generated scenarios.

hits to 0.4% and RMSE to $3{,}465$, with 93% coverage.

The multiplicative lognormal bias correction reduces NB bias from 10.9% to -0.6% under NB(n)-

Table S22: Simulation 2: recovery of γ across scenarios and estimators. Median, mean, and the share of replications where $\hat{\gamma}$ hits the upper boundary (≥ 0.49). True $\gamma = 0.005$.

Model	Median	Mean	Q75	% boundary	% > 0.1
NB(n)-NB(m)					
NB	0.0048	0.0111	0.0102	0.1	1.2
NB (c)	0.0048	0.0113	0.0102	0.1	1.2
NLS	0.0386	0.1652	0.3558	3.8	45.3
OLS	0.0099	0.0678	0.0322	6.2	13.8
Poisson	0.0038	0.0545	0.0234	0.7	14.2
Poisson (c)	0.0039	0.0581	0.0241	1.9	14.4
NB(n)-Po(m)					
NB	0.0050	0.0063	0.0061	0.1	0.2
NB (c)	0.0050	0.0053	0.0061	0.0	0.0
NLS	0.0050	0.0058	0.0066	0.0	0.1
OLS	0.0070	0.0153	0.0099	0.7	1.7
Poisson	0.0050	0.0053	0.0061	0.0	0.0
Poisson (c)	0.0050	0.0053	0.0061	0.0	0.0
Po(n)-NB(m)					
NB	0.0045	0.0155	0.0112	0.2	2.9
NB (c)	0.0045	0.0156	0.0112	0.3	2.8
NLS	0.2770	0.2389	0.3602	7.7	73.0
OLS	0.0771	0.2112	0.4526	24.1	46.4
Poisson	0.0058	0.0619	0.0420	0.9	20.9
Poisson (c)	0.0042	0.0480	0.0224	2.5	12.0
Po(n)-Po(m)					
NB	0.0060	0.1210	0.3063	16.2	26.2
NB (c)	0.0048	0.0083	0.0076	0.4	0.6
NLS	0.0084	0.0715	0.1963	0.1	31.8
OLS	0.0358	0.1422	0.2865	17.8	27.7
Poisson	0.0048	0.0073	0.0076	0.0	0.7
Poisson (c)	0.0048	0.0061	0.0075	0.0	0.1

Note: boundary = $\hat{\gamma} \geq 0.49$ (upper optimisation bound). The unconstrained NB hits this boundary in 16% of Po(n)-Po(m) replications, inflating $\hat{\gamma}$ by a factor of 24 relative to the true value.

NB(m) and from 6.9% to -4.7% under Po(n)-NB(m); the bootstrap median tracks the bias-corrected estimate closely.

F.4 Summary of simulation findings

The two simulation studies provide the following evidence for the empirical analysis.

1. Poisson PMLE, which requires only a correct mean specification, gives stable $\hat{\xi}$ estimates across scenarios; even under NB-generated data its coverage stays at 80–88% through wider CIs.
2. NB-MLE gains efficiency when correctly specified but introduces γ–θ confounding: θ absorbs variation otherwise attributed to γ, so the unconstrained NB overestimates γ by a factor of 24 in 16% of Poisson-generated replications. The constrained NB ($\alpha \in (0, 1)$) cuts boundary hits from 16% to 0.4% and is preferred when NB is used as a sensitivity check.
3. OLS with $\log(m+1)$ is severely biased when zeros are present (Simulation 2) and is unsuitable for sparse count data.

4. The multiplicative lognormal bias correction reduces NB bias from about 11% to under 1% under NB-generated data.
5. Bootstrap CIs are wider than the analytical (delta-method) CIs, particularly for NB. The empirical analysis uses a cluster bootstrap to account for within-country correlation.

G OLS and NLS estimation details

The OLS estimator operates on the log-linearised form of the model:

$$\log(m_{it}) = \alpha \log N_{it} + \beta \log\left(\frac{n_{it}}{N_{it}}\right) + \epsilon_{it}, \tag{11}$$

where α is the Community Anchor parameter and β is the Exposure Elasticity. The no-intercept specification follows directly from the theoretical model $\mu_{it} = N_{it}^{\alpha}(n_{it}/N_{it})^{\beta}$, which contains no free multiplicative constant.

This specification requires $m_{it} > 0$ and $n_{it} > 0$. When zeros are present, we either aggregate the data to eliminate zero cells or add a small positive constant, e.g. $m_{it} + 1$ and a shift parameter $\gamma > 0$ to the exposure ratio n_{it}/N_{it} to handle zero values. In simulation studies we verify the finite-sample performance when the data are aggregated in such a way that $m_{it} > 0$ and $n_{it} > 0$ (see Section F). As expected, the OLS approach is not well suited for sparse count data.

The nonlinear least squares estimator operates on the original scale:

$$m_{it} = N_{it}^{\alpha}\left(\frac{n_{it}}{N_{it}}\right)^{\beta} + \epsilon_{it}. \tag{12}$$

We consider this model both with and without the γ parameter, and all parameters are estimated using the Levenberg–Marquardt algorithm as implemented in the *minpack.lm* package in R. Unlike the count model estimators, NLS does not require distributional assumptions beyond a mean specification, which makes it a useful robustness check. However, in the simulation studies we find that the model yields unbiased estimates of ξ only when both n and m follow a Poisson distribution.

H Omitted derivations

H.1 OLS with unknown γ

The OLS estimator operates on the log-linearised form of the model. Given the specification $m_i = N_i^{\alpha}(\gamma + n_i/N_i)^{\beta}$, taking logarithms yields

$$\log(y_i) = \alpha \log N_i + \beta \log(\gamma + n_i/N_i) + \varepsilon_i,$$

where $y_i = m_i$ when all counts are positive, and $y_i = m_i + 1$ when any $m_i = 0$ (to avoid $\log(0)$). The `uncounted` package automatically selects the appropriate transformation based on the data.

Since γ enters the model non-linearly, estimation proceeds via concentrated (profile) least squares. For a fixed value of γ, the model is linear in (α, β) and can be estimated by ordinary least squares. Define the profile sum of squared errors as

$$S(\gamma) = \sum_{i=1}^{I} \Big[\log(y_i) - \hat{\alpha}(\gamma) \log N_i - \hat{\beta}(\gamma) \log(\gamma + n_i/N_i)\Big]^2,$$

where $\hat{\alpha}(\gamma)$ and $\hat{\beta}(\gamma)$ denote the OLS coefficients conditional on γ. The estimator minimises $S(\gamma)$ over the interval $\gamma \in (10^{-10}, 0.5)$ using a multi-start grid search followed by Brent's method, yielding $\hat{\gamma}$. The final estimates $\hat{\alpha}$ and $\hat{\beta}$ are then obtained from the OLS regression evaluated at $\hat{\gamma}$.

H.2 Constrained estimation via logit reparametrisation

The theoretical model requires $\alpha \in (0, 1)$ and $\beta > 0$. The unconstrained estimators (Poisson PML, NB) estimate these parameters on $\mathbb{R}$ and do not enforce the restrictions. We impose them through link functions: the logit for α and the log for β.

For observation i with covariate vector $\mathbf{x}_i$, define the linear predictor $\eta_i = \mathbf{x}_i'\boldsymbol{\eta}$ and set

$$\alpha_i = \text{logit}^{-1}(\eta_i) = \frac{1}{1 + \exp(-\eta_i)}, \qquad \beta = \exp(\zeta), \tag{13}$$

where $\boldsymbol{\eta}$ and ζ are the parameters to be estimated (on $\mathbb{R}$). The log conditional mean becomes

$$\log(\mu_i) = \text{logit}^{-1}(\mathbf{x}_i'\boldsymbol{\eta}) \cdot \log N_i + \exp(\zeta) \cdot \log\left(\gamma + \frac{n_i}{N_i}\right). \tag{14}$$

The logit link makes $\log(\mu_i)$ non-linear in $\boldsymbol{\eta}$ even under the Poisson model, so estimation requires numerical optimisation of the (pseudo) log-likelihood with respect to $(\boldsymbol{\eta}, \zeta, \gamma)$. Starting values are obtained from the unconstrained fit: $\boldsymbol{\eta}_0 = \text{logit}(\hat{\boldsymbol{\alpha}}_{\text{unc}})$, $\zeta_0 = \log(\hat{\beta}_{\text{unc}})$.

Coefficients are reported on the link scale ($\boldsymbol{\eta}$, ζ). Response-scale values $\alpha_i = \text{logit}^{-1}(\mathbf{x}_i'\hat{\boldsymbol{\eta}})$ and $\beta = \exp(\hat{\zeta})$ are obtained by applying the inverse link. Standard errors on the response scale follow

from the delta method:

$$\text{Var}(\hat{\alpha}_i) \approx [\hat{\alpha}_i(1-\hat{\alpha}_i)]^2 \cdot \mathbf{x}_i' \mathbf{V}_\eta \mathbf{x}_i, \tag{15}$$

where $\mathbf{V}_\eta$ is the variance-covariance matrix of $\hat{\boldsymbol{\eta}}$ (model-based or HC-robust).

For the population size estimator $\hat{\xi} = \sum_i N_i^{\hat{\alpha}_i}$, the gradient with respect to $\boldsymbol{\eta}$ is

$$\frac{\partial \xi}{\partial \eta_j} = \sum_i N_i^{\alpha_i} \ln(N_i) \cdot \alpha_i(1-\alpha_i) \cdot x_{ij}. \tag{16}$$

Confidence intervals for ξ are constructed by forming a Wald interval on the $\boldsymbol{\eta}$ scale and mapping it through the monotone transformation $g(\boldsymbol{\eta}) = \sum_i N_i^{\text{logit}^{-1}(\mathbf{x}_i'\boldsymbol{\eta})}$.

The multiplicative lognormal bias correction (9) in the main text does not apply under the logit link because $N_i^{\text{logit}^{-1}(\eta_i)}$ is not an exponential function of η_i. We use the second-order Taylor correction instead:

$$\hat{\xi}^{\text{BC}} = \hat{\xi} - \frac{1}{2} \sum_{i=1}^{I} N_i^{\hat{\alpha}_i} [\ln(N_i)]^2 \cdot \text{Var}(\hat{\alpha}_i), \tag{17}$$

where $\text{Var}(\hat{\alpha}_i)$ includes the delta-method adjustment (15). This correction can be positive or negative depending on α_i.

H.3 Bias correction for $\hat{\xi}_t$

Since $g(\alpha) = \sum_{i=1}^{I} N_{it}^{\alpha}$ is nonlinear in α, the plug-in estimator $\hat{\xi}_t = g(\hat{\alpha})$ is biased. Moreover, g is convex for $N_{it} > 1$, so Jensen's inequality implies $\text{E}[\hat{\xi}_t] \geq \xi_t$ when $\hat{\alpha}$ is unbiased.

A Taylor expansion of g around the true α gives:

$$\text{Bias}[\hat{\xi}_t] \approx g'(\alpha) \cdot \text{Bias}[\hat{\alpha}] + \frac{1}{2} g''(\alpha) \cdot \left(\text{Var}[\hat{\alpha}] + \text{Bias}[\hat{\alpha}]^2 \right), \tag{18}$$

where $g'(\alpha) = \sum_i N_i^{\alpha} \ln(N_i)$ and $g''(\alpha) = \sum_i N_i^{\alpha} [\ln(N_i)]^2$. Assuming $\hat{\alpha}$ is unbiased (consistent under correct mean specification for Poisson PML), the first-order term vanishes and the Taylor bias correction becomes:

$$\hat{\xi}_t^{\text{Taylor}} = \hat{\xi}_t - \frac{1}{2} \sum_{i=1}^{I} N_{it}^{\hat{\alpha}_i} [\ln(N_{it})]^2 \cdot \mathbf{x}_i' \hat{\mathbf{V}}_{\text{model}} \mathbf{x}_i,$$

where $\mathbf{x}_i$ is the i-th row of the α design matrix and $\hat{\mathbf{V}}_{\text{model}}$ is the model-based (homoscedastic) variance of $\hat{\alpha}$. This approximation has two drawbacks: (i) it can produce negative bias-corrected

estimates when variance is large and $\sum N_i^{\alpha}(\ln N_i)^2$ is large relative to $\hat{\xi}$; (ii) approximation error grows for heavy-tailed N distributions. In simulations, the Taylor correction reduces bias by about 88%.

Since $N_i^{\hat{\alpha}} = \exp(\hat{\alpha} \ln N_i)$ and $\hat{\alpha} \sim N(\alpha, V)$, the moment generating function of the normal distribution gives $\mathrm{E}[N_i^{\hat{\alpha}}] = N_i^{\alpha} \exp(\frac{1}{2}(\ln N_i)^2 V)$. Inverting this yields:

$$\hat{\xi}_t^{\mathrm{BC}} = \sum_{i=1}^{I} N_{it}^{\hat{\alpha}} \exp\left(-\frac{1}{2}(\ln N_{it})^2 \, \mathbf{x}_i' \hat{\mathbf{V}} \mathbf{x}_i\right).$$

This estimator is exact under normality of $\hat{\alpha}$, always positive, and in simulations reduces bias by about 94%. The `uncounted` package uses this form as the default.

Both corrections use the model-based variance rather than HC-robust variance, because the latter can be inflated by high-leverage observations (e.g., Ukraine with $N > 400{,}000$), leading to overcorrection. The model-based variance is computed as $\hat{\mathbf{V}}_{\text{model}} = \hat{\sigma}^2(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}$ for OLS/NLS, or $(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}$ for Poisson/NB (where $\mathbf{W}$ is the working weight matrix).

For constrained models, the delta method accounts for the logit link (15) and the Taylor correction (17) replaces the lognormal form; see Section H.2.